SPEAR3 RF Station

Legacy System Interlock Architecture

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This document traces every interlock path in the *legacy* SPEAR3 RF station: what trips, how fast, through which hardware, and what the software can and cannot see. It is the reference against which the upgraded protection chain in Doc 0 is justified. Where the two disagree, the upgrade document states the intended future behavior and this document states the present one. The Open Items register lists the points that still require field verification.

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Open Items — Still To Be Verified

Every finding in this document is traced to a source file or drawing. The items below are the ones that were *inferred* from that material and have not yet been confirmed against the installed hardware. Each is tagged **[O*n*]** at the point in the text where it is relied upon.

ID	Item	Required Evidence	Needed Before	Page
O1	Slot 5 physical module	Confirm what is physically installed in VXI slot 5 of the SRF1 crate. The crate template labels it ”MPS Shutoff” and the CF2 database is loaded as a PEP-II artifact.	VXI crate retirement	p. 19
O2	RF MPS PLC ↔ Fast IC and panel wiring	Confirm whether any hardwired line runs from the PLC-5 to the Fast Interlock Chassis besides the AB summary and AB heartbeat that pass through the RF Local Display Panel, and confirm the J3 pin order, which is reversed between SD-340-311-01 and WD-340-330-02 [R27], [R30].	Interface I/O freeze	p. 27
O3	Fast IC → AIM connection	Identify the physical cable and connector carrying the Fast IC status word, and whether it runs over the VXI backplane or an external cable.	Protection acceptance	p. 12
O4	RF Local Display Panel logic	Read out the equations of the panel’s GAL22V10 logic device, which forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort, and trace the endpoint of its beam-abort output (rear BNC J6) in the SPEAR MPS [R29].	Interface Chassis logic design	p. 11
O5	Remote I/O word/bit map	Cross-check the assignments derived from [R9] against the deployed SLC-500 ladder program.	Controller replacement	p. 30
O6	Arc sensor count	Confirm the installed sensor count and locations against the six locations and eleven total sensors quoted for the upgrade.	Arc detection changes	p. 13

Background

This document focuses on the legacy interlock system design, providing a deeper and more detailed description of the interlock architecture within the broader SPEAR3 RF system. It is part of the overall system architecture defined in [R1] — SPEAR3 RF System Legacy Architecture, which serves as the primary system-level reference. This document extends and elaborates on the interlock-related content presented there. Relevant sections in [R1]:

- Doc L, *LLRF Controller (VXI System)* — ARC Interlock Module (AIM)
- Doc L, *HVPS Control System (B118)* — SLC-500 PLC
- Doc L, *Personnel Protection System (PPS) Interface*
- Doc L, *RF Machine Protection System (MPS)*
- Doc L, *Interlock Architecture and Signal Chain*

All source code and EPICS database files cited in this document are listed with their reference numbers in Appendix B.

How to read the latency figures in this document. The response times quoted throughout (Fast Interlock Chassis $< 1\ \mu$s, AIM interrupt service $\approx 10\ \mu$s, RF MPS PLC $\approx$ 10–15 ms, SLC-500 $\approx$ 10–20 ms, SNL state machine ≈ 1 s) are *design and analysis* values derived from the cited drawings, scan-rate settings and source code. They describe the budget of each individual layer. Consistent with [R1], no measured end-to-end installed-path latency record exists for the SPEAR3 RF interlock chain, so these figures should be treated as the expected order of magnitude rather than as commissioning data.

1 Architecture Overview

1.1 The Five Actors

The SPEAR3 RF interlock system involves five distinct hardware/software actors. They are **not redundant layers of the same function** — each addresses a different threat class at a different speed:

Actor	Speed	Location	What It Protects Against
Fast Interlock Chassis (340-308)	< 1 μs	B132	Arc breakdown, reflections exceeding cavity/klystron ratings
AIM Module (VXI Slot 12)	< 1 μs (HW) / ~10 μs (ISR)	VXI Crate	Digitizes arc channel status from Fast IC; asserts RF_FAULT on VXI backplane; forces the beam abort; captures fault waveforms; drives control outputs (HVPS_On, Filament, Solenoid) through the Fast IC to the RF Local Display Panel
RF MPS PLC (PLC-5/1771)	~10 ms	B132	Collector overpower, vacuum excursion, secondary arc, cooling
SLC-500 HVPS PLC (RIO adapter 1)	~20 ms	B118	HVPS internal faults: oil, transformer, crowbar, overvoltage, arc in HV tank
SNL State Machine (`rf_states.st,v`)	~1 s	VXI CPU (VxWorks)	Orderly shutdown sequencing, fault recording, recovery coordination

1.2 Signal Flow Diagram

Three fault sources feed five actors, as shown in Figure 1. Hardware paths (top) are independent of software; the EPICS alarm tree (bottom) is the single software trip wire to the SNL state machine.

Speed summary by path:

Path	Actor	Hardware action	EPICS notification	SNL response
Arc / RF overpow. → Fast IC	Fast IC 340-308	< 1 μs (RF cut through the AIM; SCR ENABLE and CROWBAR through the AIM commands and the RF Local Display Panel logic)	~50–500 μs (AIM ISR → VXI:LTCH)	~1 s (orderly HVPS shutdown)

Path	Actor	Hardware action	EPICS notification	SNL response
MPS PLC Path A → panel logic and AIM	RF MPS PLC + RF Local Display Panel + AIM	~10–15 ms (HVPS kill through SCR ENABLE for collector and waveguide pressure faults; RF cut through the AB summary)	~1 s (Remote I/O Path B)	~1 s (orderly HVPS shutdown, HVPS already dead)
MPS PLC Path C → Slot 5	RF MPS PLC + Slot 5	~10–15 ms (RF cut only; no immediate hardware HVPS kill)	~50–500 μs (AIM ISR → VXI:LTCH → STNOFF)	~6 s (orderly HVPS shutdown via s_go_off)
SPEAR MPS permit / orbit interlock → Slot 5	VXI Slot 5	~5–15 ms (RF cut only; no *direct* hardware HVPS kill from Slot 5; but see cascade note)	~50–500 μs (AIM ISR → VXI:LTCH → STNOFF)	~6 s (SNL orderly shutdown) — **but in practice ~20–30 ms** via collector overpower cascade → MPS PLC Path A → Fast IC (see §5)
HVPS controller trip	SLC-500 HVPS PLC	~10–20 ms (24 V fault/enable removed from the trigger boards)	~100 ms–1 s (Remote I/O → EPICS)	~1 s (orderly RF shutdown, HVPS already off)

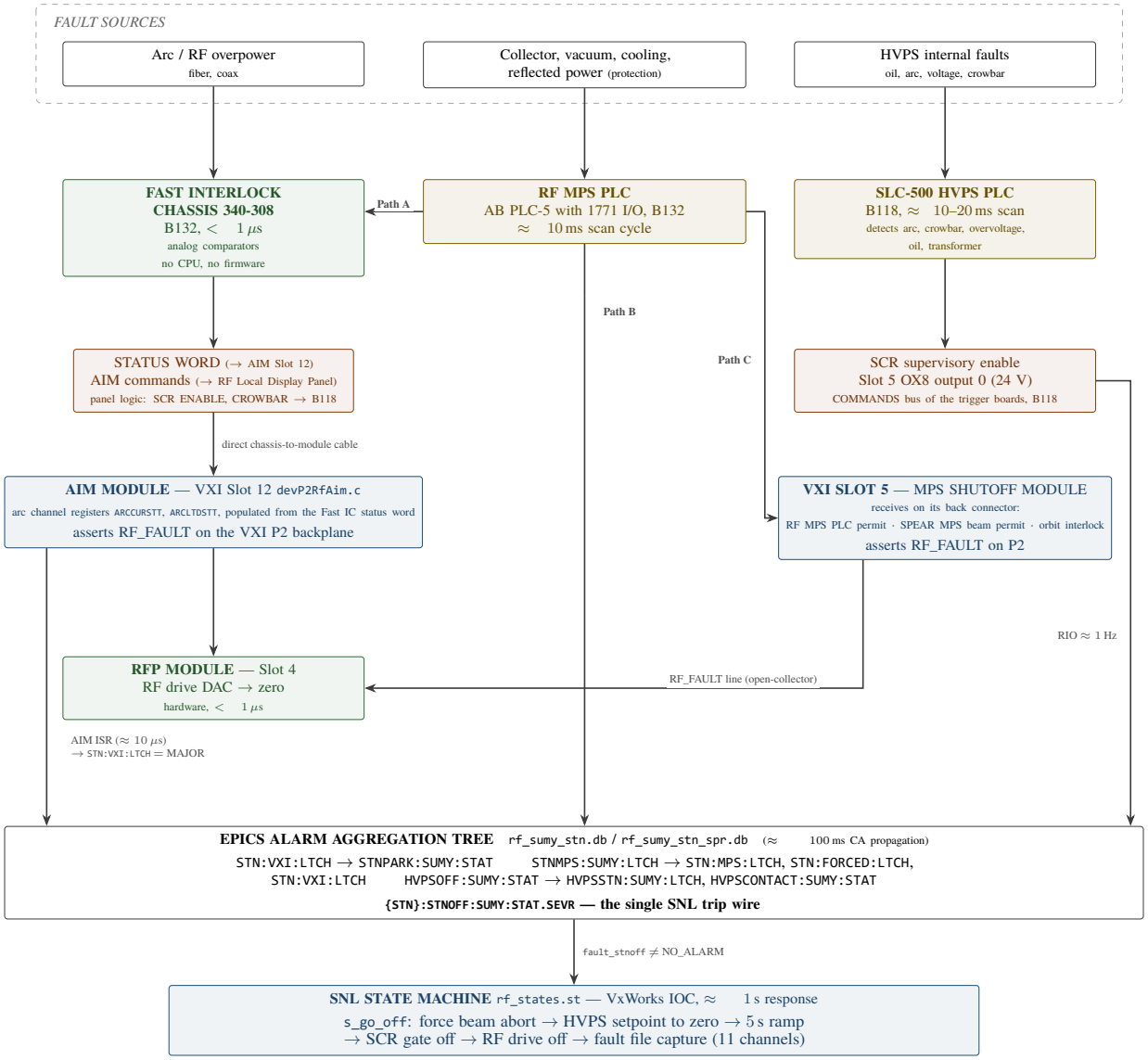


Figure 1: Legacy interlock signal flow. Three fault sources feed five actors. The hardware paths across the top are independent of software; the EPICS alarm tree at the bottom is the single software trip wire into the SNL state machine. The RF MPS PLC acts on three paths: **Path A** sends its collector and waveguide pressure summaries to the logic of the RF Local Display Panel (not drawn), which removes SCR ENABLE, and its AB summary over the panel's fibers to the Fast Interlock Chassis and the AIM; **Path B** clears the Remote I/O permit bit, setting `STN:MPS:LTCH` to MAJOR in the alarm tree; **Path C** drives VXI Slot 5, which asserts RF_FAULT on the backplane as a redundant RF cut. The three HVPS fibers run between the RF Local Display Panel in B132 and the HVPS controller in B118.

2 Fast Interlock Chassis 340-308

Sources: [R1], [R6], [R19].

2.1 Function

The Fast Interlock Chassis is a pure analog hardware circuit — no software, no firmware, no CPU in the trip path. It provides sub-microsecond RF protection that is completely independent of all PLCs and computers.

Physical location: B132 electronics rack, right below the VXI crate.

2.2 Inputs

Input Type	Source	Connection	Notes
Arc detection (fiber optic)	4× cavity waveguide window sensors	Fiber optic receivers on chassis	Raw optical arc pulses → analog comparators
Arc detection (fiber optic)	Klystron output window sensor	Fiber optic receiver	
Arc detection (fiber optic)	Main circulator sensor	Fiber optic receiver	
RF reflected power	Detector diodes at cavity and waveguide junctions	Coaxial cable, analog voltage	Compared against DC threshold
RF forward power	Detector diodes (klystron output monitor)	Coaxial cable, J16 ”DETECTED KLYSTRON POWER”	
HVPS status (`HVPSON` in AIM `FISTAT`, bit 6)	Not documented	—	The HVPS STATUS fiber ends on the RF Local Display Panel, where it is received as the HVPS beam abort, not on this chassis [R26], [R27]; the source of the `FISTAT` bit is not documented. The AIM uses the bit to gate arc voltage readback (arc peaks not read when `HVPSON=0`).
AB summary and AB heartbeat	RF MPS PLC-5, through the RF Local Display Panel	Two fibers from the panel; passed to the AIM on J15	AB summary fault reaches the AIM (Path A) [R26]

Signal routing correction: The SPEAR MPS beam permit and orbit interlock signals are **not** inputs to the Fast Interlock Chassis. They are wired to the back connector of the **VXI Slot 5 MPS Shutoff module**. When either permit is removed, Slot 5 asserts `RF_FAULT` on the VXI P2 backplane, cutting RF drive. See §4.2 for the complete Slot 5 input description.

2.3 Outputs

Output	Destination	Latency	Mechanism
AIM commands (six)	RF Local Display Panel (B132)	< 1 μs	FILAMENT_ON, SOLENOID_ON, HVPS_ON, BEAM_ABORT, FILAMENT_TIMEOUT and FAULT_RESET from the AIM, inverted and sent as fibers [R26]
AB summary, AB heartbeat	AIM Module (VXI Slot 12)	~1 μs	From the panel's fibers, on J15
Status word	AIM Module (VXI Slot 12)	~1 μs	Direct chassis-to-module connection

SCR ENABLE and KLYSTRON CROWBAR are not outputs of this chassis. They are formed by a programmable logic device (GAL22V10) in the RF Local Display Panel 340-310 from the HVPS on request and the VXI beam abort (the AIM's commands, through this chassis), the HVPS STATUS fiber, the main and bucking solenoid faults and the PLC-5's collector and waveguide pressure summaries. They leave the panel on fibers to the trigger interconnect boards of the HVPS controller in B118; STATUS returns from there to the panel. The same logic drives the beam abort [R26], [R27], [R29], [R30]. The equations of the logic are not documented. **[O4]**

These latencies are signal-propagation times, not fault-clearing times. The distinction matters when reasoning about fault energy:

Stage	Time
Fiber command changes (AIM, Fast IC, panel logic)	**μs scale** (not measured)
Crowbar thyristor begins conducting	$\approx$ **10 μs** after the trigger
Primary AC current interrupted	**4–8 ms**
Vacuum contactor opens (PPS Chain 1)	$\approx$ **200 ms** worst case

Removing SCR gate drive does **not** stop conduction immediately — the thyristors continue to the next current zero. The crowbar is what bounds energy into the klystron: **< 5 J with it, < 40 J without** (SLAC-PUB-7591).

2.4 Front Panel

The front panel of the Fast Interlock Chassis (SLAC drawing 340-308; photographed in Doc L) shows:

- **FAULT RESET** button — latches cannot be cleared without this
- **LAMENT TIMEOUT** indicator — klystron filament interlock
- **BEAM ABORT** indicator/control
- **TEST PORT** 12-channel test port for arc threshold adjustment
- **INPUT MONITOR** — CH.1 through CH.12 input monitor test points
- **HVPS ON**, **SOLENOID ON**, **FILAMENT ON** status indicators
- **J16 DETECTED KLYSTRON POWER** coaxial input

2.5 Trip Mechanism

Each arc detection channel has an adjustable current threshold set by AIM configurable with software. When a fiber-optic arc pulse arrives (arc sensors generate an optical burst on breakdown):

1. Analog comparator fires
2. Fault latch is set (latching relay or SR flip-flop — requires hardware FAULT RESET to clear)

3. The status word to the AIM changes; the AIM latches the arc and asserts `RF_FAULT`
4. The AIM's six command signals are set by its register bits and by its on-board fault logic [R31]; when they drop the HVPS on request or force the beam abort, the RF Local Display Panel logic removes SCR ENABLE and, as its equations specify, fires the crowbar (O4)

For reflected power, the analog input voltage from the RF detectors is compared to a DC threshold. Exceeding the threshold fires the same trip mechanism.

Source: [R1] (interlock architecture and signal chain); [R21].

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3 AIM Module (VXI Slot 12)

Sources: [R3], [R6], [R7], [R14], [R21].

3.1 Role Clarification — AIM vs. Fast Interlock Chassis

The AIM module (drawing 340-307) and the Fast Interlock Chassis (drawing 340-308) are companion units designed as a pair. Their sequential SLAC drawing numbers reflect this: 340-307 is the VXI digital module; 340-308 is the external analog chassis.

Fast Interlock Chassis 340-308 is the **analog front-end**:

- Directly receives fiber-optic arc sensor pulses from cavity windows, klystron output, and circulator
- Performs sub-microsecond analog threshold comparisons (no CPU, no firmware, no software in the trip path)
- Passes the AIM's six command signals to the RF Local Display Panel as fibers and returns the PLC-5's AB summary and heartbeat to the AIM; SCR ENABLE and KLYSTRON CROWBAR are formed by the panel (§2.3)
- Sends the resulting arc-channel status word to the AIM module via a direct chassis-to-module cable

AIM module 340-307 is the **VXI digital companion**:

- Receives arc-channel status word from Fast IC via direct hardware connection; the physical cable and connector are not identified [O3]
- Its 12-channel arc registers (`ARCCURSTT`, `ARCLTDSTT`) digitally represent the Fast IC's per-channel arc detector states
- **Provides the VXI configuration interface for Fast IC comparator thresholds**: writing to the AIM's `TRIPLVL` registers (A24 address space) sets the arc detection threshold for each channel; the Fast IC reads these values from the AIM via their direct link and applies them to its analog comparators. Enforced range: 10–255 (MINTRIPLVL / MAXTRIPLVL in `devP2RfAim.c`).
- Asserts `RF_FAULT` on the VXI P2 backplane (sub-μs) to cut RF drive from the RFP module
- Generates the ISR $\rightarrow$ EPICS alarm chain notification (~10 μs)
- Drives six control signals, which the Fast Interlock Chassis converts to fibers for the RF Local Display Panel: station online request, filament and focus request, HVPS on request, VXI beam abort, filament timeout bypass and reset [R26], [R31]
- Forces the beam abort and resets it; the beam abort reaches the SPEAR MPS through the panel's beam-abort output (§3.8)
- **Captures arc-channel history waveforms** via on-module 512 KB ADC ring buffer (12-ch AIM `HISBUF`): the Fast IC's fast ADC continuously samples all arc channel voltages and the HVPS voltage into this buffer; on fault the buffer freezes and is read back

- Captures RF/IQA fault waveforms at the SNL level (11 channels → `/dat/FAULT*_N` files)

Key distinction: The AIM IS correctly named as an arc detector — it registers and latches per-channel arc events and asserts the VXI-level RF protection. However, it does **not** process raw fiber-optic arc pulses directly. All raw optical signal processing is done by the Fast IC 340-308. The AIM's arc detection is the digital representation and VXI-backplane consequence of what the Fast IC detected.

3.2 Hardware Identity

- **Module type**: SLAC PEP-II RF custom VXI module — "Arc detector / Interlock Module" (AIM), drawing 340-307
- **VXI slot**: 12 (last slot, rightmost in crate)
- **Device support driver**: `devP2RfAim.c` (1,982 lines)
- **Primary EPICS record**: `{STN}:STN:AIM:MODU` (type `p2RfAim`)
- **DAS instruction files**: `/tbl/aimDas0.inst`, `/tbl/aimDas1.inst`
- **History file**: `/dat/aimHist.dat`

3.3 Arc Detection (12 Channels)

The AIM module has 12 independently configurable arc detection channels. Each channel has:

- **Enable/disable** bit (`ARCENBSTT` register, `MODU.AFES` field)
- **Threshold** setting (8-bit value, range 10–255): configured via EPICS writing to the AIM's `TRIPLVL` registers (A24 offset `0x0020` for ch 0, successive offsets per channel). The device driver (`devP2RfAim.c`) writes the value to the AIM VXI register; the Fast IC reads it via their direct hardware link and applies it to its analog comparator for that channel. The old method of adjusting thresholds via the Fast IC front panel is superseded by this software interface.
- **Current status** (`ARCCURSTT`, `MODU.ACST`) — real-time arc presence (populated from Fast IC status word)
- **Latched status** (`ARCLTDSTT`, `MODU.ALST`) — fault held since last reset (populated from Fast IC status word)

> **Note on channel count**: The legacy AIM supports 12 hardware channels. The SPEAR3 deployment uses 6 sensor locations (4 cavity windows + 1 klystron window + 1 circulator), provisioned with up to 11 total sensors including spares. The authoritative upgrade count is in Doc 0; the installed count has not been confirmed by inspection. **[O6]**

3.4 Hardware Trip Path — RF_FAULT Backplane Line

This is the fastest path, entirely in hardware with no software involvement (Figure 2):

The `RF_FAULT` line is **open-collector**: any module on the VXI P2 bus can assert it. The RFP module's reaction is purely analog — no DSP, no SNL, no EPICS involved.

3.5 ISR Path — Software Notification (~10 μs)

After the hardware trip fires:

```
AimIsr() {
    read FAST INTERLOCK STATUS REGISTER  // latched, not ISR register (Sass 2004 fix)
    update rec->istt
    board->intPrc++
    scanOnce(rec)   // schedules AIM:MODU for EPICS processing
}
```

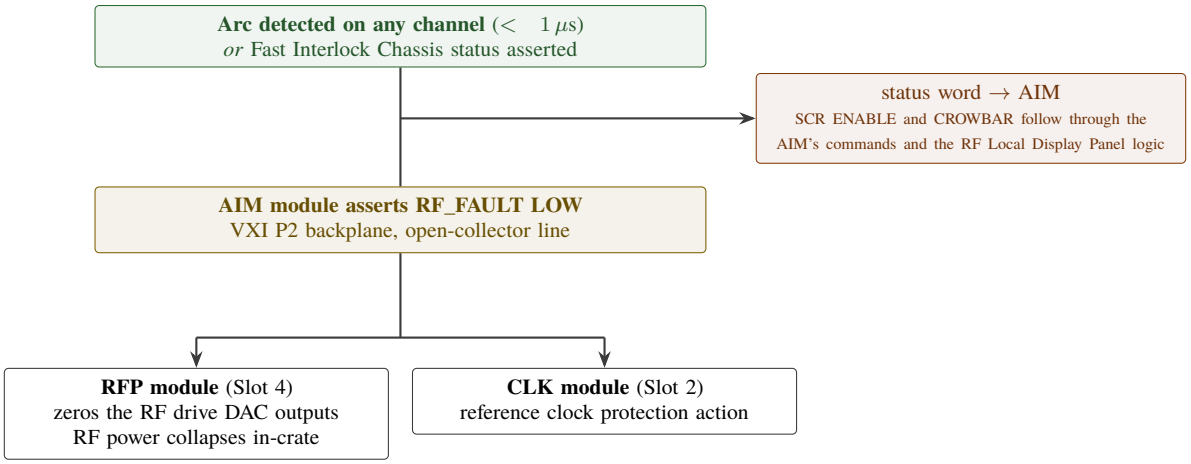


Figure 2: Fast Interlock Chassis trip mechanism. Pure analog comparators with no CPU in the path.

Critical 2004 fix (Sass, `rf_interlock_vxi.db` rev 1.2): Prior to January 2004, the VXI interlock records read the AIM **interrupt status register** (ISR). Because EPICS processed these records continuously, the ISR bits self-cleared on every read — a latched fault could be invisible to the alarm system because it was cleared before alarm propagation. The fix changed to reading the AIM **fast interlock status/latch register**, which holds bits until the operator presses the hardware FAULT RESET button on the interlock chassis front panel. This was a significant reliability fix.

The record `{STN}:STN:AIM:MODU` then processes via a three-stage fanout (`FANO1` → `FANO2` → `FANO3`) that updates all derivative records.

3.6 AIM Status PV Map

PV	AIM Register	Content
`{STN}:STN:AIM:ARCCURSTT`	`MODU.ACST`	mbbiDirect — 12-bit arc current status word (one bit per channel)
`{STN}:STN:AIM:ARCLTDSTT`	`MODU.ALST`	mbbiDirect — 12-bit arc latched status (holds until FAULT RESET)
`{STN}:STN:AIM:ARCENBSTT`	`MODU.AFES`	mbbiDirect — 12-bit arc enable bitmap
`{STN}:STN:AIM:FSTFLT`	`MODU.FFLT`	mbbiDirect — fast fault status from Fast IC
`{STN}:STN:AIM:FSTINTSTT`	`MODU.FSTT`	mbbiDirect — fast interlock state (daisy chain)
`{STN}:STN:AIM:INTSTATE`	`MODU.ISTT`	mbbiDirect — overall AIM interlock state
`{STN}:STN:AIM:INTACK`	`MODU.IACK`	mbboDirect — interlock acknowledge

These feed into `rf_interlock_vxi.db`:

```
STN:STN:AIM:LTCH.BN  →  STN:STN:VXI:LTCH  (bi, MAJOR alarm)
    → STN:STNMPS:SUMY:LTCH.INPC
```

3.7 AIM Control Output Signals (Fiber Optic Outputs)

The AIM module drives **6 control output signals** via the `FICTRL` (Fast Interlock Control) register (A16 address `0x20`). They are set by the SNL state machine and by the AIM's on-board fault logic [R31]. They leave the AIM on J3, pass through the Fast Interlock Chassis (J15), which inverts them and converts them to light, and reach the RF Local Display Panel as six fibers [R26]:

Signal Name	EPICS PV	AIM FICTRL bit	Destination	Function
Klystron Filament On	`{STN}:STN:AIM:FILAMENT`	bit 2 (`MODU.FOO`)	RF Local Display Panel (station online request), then the PLC-5	Enables klystron filament heater; required before HVPS startup
Solenoid On	`{STN}:STN:AIM:SOLENOID`	bit 0 (`MODU.SOO`)	RF Local Display Panel (filament and focus request), then the PLC-5	Enables solenoid; required before HVPS startup
HVPS On (ai-mon)	`{STN}:STN:AIM:MODU.HVPS`	bit 1 (`MODU.HVPS`)	RF Local Display Panel logic (HVPS on request)	**Hardware enable gate** — AIM holds LOW at startup; SCR ENABLE cannot be lit until IOC asserts it in `HVPSONSUB`
Force Beam Abort (fba)	`{STN}:STN:AIM:FRCBMABT`	bit 6 (`MODU.FBA`)	RF Local Display Panel (VXI beam abort), then its beam-abort output	Station not ready for beam; asserted in `s_go_off` Step 1
Fault Reset	`{STN}:STN:AIM:MODU.RSTF`	bit 5 (`MODU.RSTF`)	Fast IC arc latches; RF Local Display Panel reset	Clears arc latches in Fast IC and AIM (equivalent to pressing front-panel FAULT RESET button) and the panel's latches
Filament Timeout	`{STN}:STN:AIM:MODU.FTOR`	bit 3 (`MODU.FTOR`)	RF Local Display Panel (filament timeout bypass), then the PLC-5 as a 24 V request	Filament timer bypass; not commonly used in normal SPEAR3 operation

Note: `Forced_Fault` (bit 4, `MODU.FF`, PV `{STN}:STN:AIM:FRCFLT`) is a test input of the Fast IC, not one of the six fibers, and `Beam_Abort_Reset` (bit 7) resets the AIM's beam-abort latch without a fiber of its own [R26], [R31].

Strobed signals (HVPS, FILTMOOVRD, FLTRESET, RSTBMABT) are written high then immediately re-cleared in the same processing cycle to avoid latching; latching signals (SOLENOID, FILAMENT, FRCDFLT, FRCBMABT) hold their state.

3.8 BATS — Beam Abort Trip Signal

The state machine forces a beam abort (`fba`) whenever it turns the station off or enters TUNE (`TUNESUB`), and resets it when the station reaches ON or PARK with no faults [R2]. The forced abort leaves the AIM as the VXI beam abort fiber, through the Fast Interlock Chassis, to the RF Local Display Panel. The panel's logic

combines it with the HVPS STATUS fiber (the HVPS beam abort) and the main and bucking solenoid faults, latches the result until reset and drives a rear-panel BNC output (J6) [R26], [R29]. The signal therefore tells the SPEAR MPS that the RF station is not ready for beam. The endpoint of J6 and the action taken at the SPEAR MPS are not on any RF drawing (O4).

BATS reset sequence (required before restarting):

```
/* RESET_BMABTSUB macro in rf_states.st */
if (fault_stnoff == NO_ALARM) {
    fba = 0;        pvPut(fba)   // de-assert Force Beam Abort
    pvPut(rba)                   // Reset Beam Abort (RBA → MODU.RBA)
    pvPut(rstf)                  // Reset Faults (RSTF → MODU.RSTF --- clears arc latches)
}
```

This three-write sequence is only executed when `fault_stnoff == NO_ALARM` — all fault conditions must be clear first.

3.9 Fault Capture — Two Independent Systems

There are **two completely separate** capture systems that activate on fault. They capture different signals by different mechanisms at different speeds.

3.9.1 System A — AIM Hardware History Buffer (12-channel AIM only)

The 12-channel AIM module includes a 512 KB on-module ring buffer (`HISBUF`, A24 address `0x0038`, `AIM_K_HISBUFID`). The **Fast IC has an onboard fast ADC** that continuously samples all arc channel voltages and the HVPS voltage, storing them into this AIM memory. Two `FICTRL` bits control the ADC:

FICTRL bit	Name	0	1
bit 8	`ADCMUX`	HVPS voltage + all 12 arc channels (normal)	HVPS voltage only
bit 9	`ADCCTL`	Start / continue buffering (normal; no fault)	Stop buffering (freeze on fault)

Normal operation: `ADCCTL=0` and no fault → ADC continuously overwrites the ring buffer.
On fault: The fault assertion sets `ADCCTL=1` → buffer freezes at fault instant → pre-fault arc waveforms and HVPS voltage are preserved.

Buffer readout: The driver reads arc voltage peak values via the `ARCVOL` register (A24 `0x003A`) — a sequential-readout register that returns one channel's voltage per read, cycling through all channels. Similarly, `CRTVOL` (A24 `0x003C`) returns per-channel crate voltage measurements. The readout is gated: if `FISTAT.HVPSON = 0` (HVPS off), the driver skips arc voltage readback (no HV → no expected arc peaks).

> **7-channel AIM (legacy PEP-II)**: Used a different mechanism — two DAS (Data Acquisition System) chips with instruction files (`/tbl/aimDas0.inst`, `/tbl/aimDas1.inst`) that defined what registers to capture. This DAS mechanism is NOT used in the 12-channel SPEAR3 AIM. The `/dat/aimHist.dat` file listed in §3.2 is the 7-ch DAS artifact; SPEAR3 uses the HISBUF hardware buffer.

3.9.2 System B — SNL Fault File Capture (`ss rf_statesFF`)

A software-level capture triggered by `efSet(ffwrite_ef)` in the SNL `s_go_off` state at ~1 s after the hardware fault. Captures **11 RF/IQA signal channels** (not arc voltages). The `ss rf_statesFF` state set:

1. Increments fault slot number (circular buffer 1–15, `NUMFAULTS=15`)
2. Timestamps via `pvTimeStamp(hvpswdefault)` — the HVPS voltage PV's timestamp
3. Places 11 signal modules in LOAD mode
4. Reads hardware signal memory buffers from RFP, IQA modules
5. Writes to `/dat/FAULTxxx_N` files (N = fault slot 1–15)
6. Maximum wait: `MAXFFWAIT = 180` × `20 ticks` ≈ `3.6 s`

The 11 fault file channels:

Channel	File Pattern	Signal
0	`FAULTRfpSI_N`	RFP Signal I (cavity forward, I)
1	`FAULTRfpSQ_N`	RFP Signal Q (cavity forward, Q)
2	`FAULTRfpCI_N`	RFP Cavity I (cavity voltage, I)
3	`FAULTRfpCQ_N`	RFP Cavity Q (cavity voltage, Q)
4	`FAULTIqa1Amp_N`	IQA1 amplitude waveform
5	`FAULTIqa2Amp_N`	IQA2 amplitude waveform
6	`FAULTGvf_N`	GVF signal (PEP-II heritage, inactive in SPEAR3)
7	`FAULTAim_N`	AIM arc/interlock status snapshot
8	`FAULTCmbI_N` / `FAULTCmbQ_N`	Combiner I/Q (PEP-II, inactive in SPEAR3)
9	`FAULTIqa3Amp_N`	IQA3 amplitude waveform
10	*(additional)*	Additional channel added in 2003 Laznovsky revision

Source: [R2] lines 1900–2200 (`ss rf_statesFF`); [R3]; [R21]; [R6] (`AIM_A_HISBUF`, `AIM_V_ADCMUX`, `AIM_V_ADCCTL`).

3.10 Inputs

Input	Source	Connection	Notes
Arc channel status (12 channels, `ARCCURSTT`)	Fast Interlock Chassis 340-308	Direct chassis-to-module cable	Digital arc-detected status word, one bit per channel; ARCLTDSTT holds bits until FAULT RESET
Fast IC interlock chain state (`FSTINTSTT`, `MODU.FSTT`)	Fast Interlock Chassis 340-308	Direct chassis-to-module cable	Daisy-chain fast interlock state
Fast IC fast fault status (`FSTFLT`, `MODU.FFLT`)	Fast Interlock Chassis 340-308	Direct chassis-to-module cable	Aggregated fast fault indication
AIM interrupt	AIM on-board IRQ logic	VXI interrupt line	`AIM_M_ARCDETD` (arc detected), `AIM_M_VXIBPFLT` (VXI backplane fault), `AIM_M_FRCDFLTINT` (forced fault), etc.

Input	Source	Connection	Notes
AB summary and AB heartbeat	RF MPS PLC-5, through the RF Local Display Panel fibers and the Fast IC	J15 of the Fast IC to J3 of the AIM [R26]	Watchdog for the PLC-5 and its summary fault; contributes to `AIMARC:LTCH` via first-fault register
AIM_M_SOFTWARE interrupt	EPICS IOC write	VXI register write	Software-triggered interrupt (for test/forced-fault operations)

3.11 Outputs

Output	Destination	Mechanism	Notes
`RF_FAULT` assertion (open-collector LOW)	VXI P2 backplane	VXI bus	Sub-μs; causes RFP (slot 4) to zero RF drive DAC and CLK (slot 2) to react
HVPS_On fiber (`aimon`, `MODU.HVPS`)	RF Local Display Panel logic (HVPS on request $\rightarrow$ SCR ENABLE)	Through the Fast IC, then fiber	Software gate; must be explicitly asserted by SNL `HVPSONSUB`; AIM holds LOW at startup
Klystron Filament On (`MODU.FOO`)	RF Local Display Panel (station online request), then the PLC-5	Through the Fast IC, then fiber	Commanded by SNL; part of HVPS startup sequence
Solenoid On (`MODU.SOO`)	RF Local Display Panel (filament and focus request), then the PLC-5	Through the Fast IC, then fiber	Commanded by SNL; required before HVPS can be energized
Force Beam Abort (`MODU.FBA`, `fba`)	RF Local Display Panel (VXI beam abort), then its beam-abort output	Through the Fast IC, then fiber	Station not ready for beam; asserted in `s_go_off` Step 1
Reset Beam Abort (`MODU.RBA`, `rba`)	AIM beam-abort latch	AIM register write	De-asserts the forced beam abort; used in recovery `RESET_BMABTSUB`
Fault Reset (`MODU.RSTF`, `rstf`)	AIM hardware latches	AIM register write	Clears `ARCLTDSTT`, `AIMARC:LTCH`, fast interlock latches — same effect as hardware FAULT RESET button
EPICS PVs (arc status, interlock state)	EPICS IOC (VxWorks)	EPICS CA record processing	`ARCCURSTT`, `ARCLTDSTT`, `ARCENBSTT`, `FSTFLT`, `INTSTATE` $\rightarrow$ `STN:VXI:LTCH` $\rightarrow$ alarm tree

---

4 VXI Slot 5: ”MPS Shutoff” — Clarification

Sources: [R5], [R17], [R18], [R19], [R20].

4.1 What It Is

This slot requires careful distinction between hardware labeling, software configuration, and functional role.

The SPEAR3 VXI crate (SRF1) at slot 5 is labeled **"MPS Shutoff"** in the crate physical template (`crat_vxi_13slot.template,v`) [R17]. This is its functional designation in the SPEAR3 system; what is physically installed in the slot has not been confirmed. [O1]

The IOC software (`rf_vxi_modules_All.substitutions,v`) [R18] loads `cf2.db` (the Comb Filter 2 database) for slot 5 — **this is a PEP-II inherited template artifact**. The `devP2RfCf2.c` driver (2,970 lines) is a PEP-II-only module; the CF2 DSP functionality (comb filter, group delay equalizer, filter bank) is **not exercised in SPEAR3**.

Aspect	PEP-II	SPEAR3 SRF1
Physical slot 5 module	CF2 (Comb Filter 2) DSP	MPS Shutoff module
Software DB loaded	`cf2.db`	`cf2.db` (PEP-II template artifact)
Device driver active?	Yes — `devP2RfCf2.c`	No — physical CF2 module not installed
Functional use	RF comb filter feedback	MPS permit interface

Source: [R19] line 65; [R20] line 95.

4.2 Hardware Function

The "MPS Shutoff" module in slot 5 provides the **VXI-backplane permit interface** for machine-level and machine-protection signals. Its role is:

1. The module occupies slot 5 and has access to the VXI P2 backplane
2. It receives **three permit signals** at its back connector (rear I/O):
   - **RF MPS PLC permit** — from the RF MPS PLC (PLC-5/1771), via a hardwired contact
   - **SPEAR MPS beam permit** — from the SPEAR machine-level MPS, directly wired to the back connector
   - **Orbit interlock** — from the SPEAR3 orbit feedback system, directly wired to the back connector
3. When all permits are **present**: the module holds the `RF_PERMIT` backplane line HIGH (or asserts a permissive signal); the RFP module (slot 4) is allowed to output RF drive
4. When any permit is **removed**: the module drops `RF_PERMIT` or asserts `RF_FAULT` on the backplane; the RFP module immediately cuts output (same hardware-speed path as the arc trip)

Key routing fact: The SPEAR MPS beam permit and orbit interlock signals wire directly to the **back connector of the MPS Shutoff module (Slot 5)**. They do **not** connect to the inputs of the Fast Interlock Chassis 340-308 or of the RF Local Display Panel. This means removal of these permits cuts RF drive immediately (via VXI backplane `RF_FAULT`) but does **not** remove SCR ENABLE or fire the crowbar directly — there is no immediate hardware HVPS shutdown path through the SPEAR MPS permit or orbit interlock. However, the `RF_FAULT` assertion on the VXI backplane propagates through the EPICS alarm chain (`STN:VXI:LTCH` MAJOR → `STNPARK:SUMY:STAT` MAJOR → `STNOFF:SUMY:STAT` MAJOR → `fault_stnoff` in the SNL state machine), which drives `s_go_off` and performs an orderly HVPS shutdown approximately 6 seconds later.

4.3 Software Visibility

The MPS permit status is tracked in EPICS via `{STN}:STN:MPS:LTCH` — a binary input record reading from the AB Remote I/O communication registers (`rf_interlock.db`, `DTYP="AB-1771DCM BI"`, word `T6[WL,B]`).

This is the software-visible representation of the MPS permit held by the RF MPS PLC.

The {STN}:STN:VXI:LTCH record (from rf_interlock_vxi.db) reads from the AIM latch register bits ({STN}:STN:AIM:LTCH.BN) — not from the CF2 slot 5 module. The CF2 OVFL and status records ({STN}:STN:CF2OVFL:STAT) are loaded in the DB but are not connected to any active hardware in SPEAR3 (their DTYP="Soft Channel" reads from the inactive {STN}:STN:CF2:MODU record).

4.4 Key Conclusion

VXI Slot 5 in SPEAR3 is a permit gate at the VXI backplane level. It receives three permit signals at its back connector: the RF MPS PLC permit, the SPEAR MPS beam permit (directly wired), and the orbit interlock (directly wired). It does NOT run the CF2 DSP firmware. The cf2.db PV records loaded for slot 5 are inactive (hardware not present). Removal of any permit causes the RFP module to cut RF drive at hardware speed through the VXI backplane permit line and — via the EPICS alarm chain (STN:VXI:LTCH → STNOFF:SUMY:STAT → fault_stnoff) — the SNL state machine performs an orderly HVPS shutdown ~6 s later. The SPEAR MPS permit and orbit interlock do NOT connect to the Fast Interlock Chassis or the RF Local Display Panel: their removal produces no **immediate hardware** HVPS shutdown from Slot 5 itself (no SCR ENABLE removal, no crowbar), but HVPS IS shut down through the SNL software path. However, the RF drive cutoff invariably creates a cascade secondary trip — with drive = 0 and HVPS on, the klystron collector absorbs full cathode power, the RF MPS PLC drops its collector summary (Path A) within ~10–15 ms and the RF Local Display Panel logic removes SCR ENABLE within ~20–30 ms total. See §5 for the complete cascade physics.

4.5 Analysis: Is the MPS Shutoff Module Necessary?

The MPS PLC acts in hardware through the RF Local Display Panel (its collector and waveguide pressure summaries feed the panel logic, and its AB summary reaches the AIM over the panel and Fast IC fibers) and through a permit input of the VXI Slot 5 module. In addition, the SPEAR MPS beam permit and orbit interlock wire **directly** to the Slot 5 back connector — independently of the RF MPS PLC. When the RF MPS PLC trips, or when either direct permit is removed, the following consequences apply:

Path	Triggered By	Immediate HVPS Action	Delayed HVPS Action (SNL)	RF Drive Action
Path A — RF Local Display Panel and AIM (§6.5)	Collector or waveguide pressure summary drops; AB summary fault	**SCR ENABLE removed by the panel logic** (fiber → HVPS controller, B118, ~10–15 ms); crowbar as the logic specifies (O4)	Orderly ramp-to-zero + hvpstrig=OFF (~1–2 s, but HVPS already dead)	RF_FAULT via AIM → VXI backplane
Path C — VXI Slot 5 (RF MPS permit)	RF MPS permit removed → Slot 5 loses permit	**None** — Slot 5 has no connection to the SCR ENABLE or crowbar fibers	Orderly HVPS shutdown via SNL s_go_off (~6 s: VXI:LTCH → STNOFF → fault_stnoff)	RF_FAULT asserted on VXI P2 backplane
Path C — VXI Slot 5 (SPEAR MPS direct)	SPEAR MPS beam permit removed → Slot 5 loses permit	**None** — Slot 5 has no connection to the SCR ENABLE or crowbar fibers	Orderly HVPS shutdown via SNL s_go_off (~6 s: same alarm chain)	RF_FAULT asserted on VXI P2 backplane
Path C — VXI Slot 5 (orbit direct)	Orbit interlock trips → Slot 5 loses permit	**None** — Slot 5 has no connection to the SCR ENABLE or crowbar fibers	Orderly HVPS shutdown via SNL s_go_off (~6 s: same alarm chain)	RF_FAULT asserted on VXI P2 backplane

Assessment:

The Slot 5 RF drive kill via VXI backplane is **functionally redundant** with the RF_FAULT assertion that already results from Path A (AB summary → AIM) for an RF MPS PLC trip. Both respond to the same PLC fault, and both assert `RF_FAULT` on the VXI P2 bus at sub-millisecond to ~15 ms speed.

For the SPEAR MPS permit and orbit interlock inputs, Slot 5 is the **exclusive** RF kill path — there is no parallel Fast IC action for these signals. Their removal cuts RF drive immediately (via VXI backplane) and also triggers an orderly HVPS shutdown via the SNL state machine approximately 6 seconds later through the following alarm chain:

```
Slot 5 RF_FAULT → AIM ISR (~10 μs) → STN:VXI:LTCH (MAJOR)
    → STNPARK:SUMY:STAT (MAJOR) → STNOFF:SUMY:STAT (MAJOR)
    → fault_stnoff ≠ NO_ALARM → s_go_off (SNL rf_states.st)
        → HVPS voltage setpoint → 0 (ramp, 5 s)
        → hvpstrig = OFF  (STN:HVPSSCR:ON:CTRL = 0)
```

The key distinction between fault sources is **how quickly** the HVPS is shut down:

Fault source	Immediate HVPS hardware kill from this path	HVPS shutdown time
Arc / RF overpow. (Fast IC)	Through the AIM's commands and the panel logic, as their logic specifies (O4)	μs scale if so; otherwise through the cascade or the SNL
RF MPS PLC (Path A)	Yes for collector and waveguide pressure faults — SCR ENABLE removed by the panel logic (~10–15 ms); other faults through the cascade	~10–30 ms (hardware)
SPEAR MPS / orbit (Slot 5)	**No** — no panel input in this path from Slot 5	~6 s (SNL orderly shutdown) — **but in practice ~20–30 ms** via collector overpower cascade → Path A (see §5)

Critically, Slot 5 does **not** independently produce an immediate HVPS shutdown for its direct-permit inputs. The hardware HVPS kill (SCR ENABLE removal, and the crowbar if the logic so specifies) comes from the RF Local Display Panel logic responding to its own inputs. However, the RF drive cutoff itself creates a cascade secondary trip through the RF MPS PLC collector overpower protection: with drive = 0 and HVPS on, the klystron collector absorbs full cathode power, the PLC detects the overpower within ~10 ms and drops its collector summary (Path A), and the panel logic removes SCR ENABLE within ~20–30 ms of the initial RF cut. See §5 for the full mechanics and signal chain.

Why Slot 5 is essential — not merely retained:

Slot 5 is **architecturally necessary** for SPEAR3 RF protection:

1. **Sole integration point for SPEAR MPS and orbit interlock**: These two signals are wired only to the Slot 5 back connector. Without Slot 5, their removal would have zero effect on either RF drive or HVPS — the klystron would continue transmitting into the cavity regardless of the machine beam-permit status or orbit feedback state. There is no alternative wiring path, no RIO register, and no EPICS record that would carry these signals to any other protective device.
2. **Provides the only VXI-backplane RF kill for machine-level permits**: The Fast IC, the RF Local Display Panel and the SLC-500 HVPS PLC have no knowledge of SPEAR MPS beam permit or orbit interlock state. Only Slot 5 bridges these machine-level signals into the RF station protection architecture.

3. **Defense-in-depth for RF MPS PLC trips (Slot 5 redundant with Path A for RF cut)**: For an RF MPS PLC trip, Slot 5 provides a redundant direct-hardware RF kill via VXI backplane that is independent of the AIM module's health. If the AIM failed to assert `RF_FAULT` after an AB summary fault, Slot 5 would still cut RF drive.
4. **Carried forward from PEP-II architecture**: The CF2 slot served a different function in PEP-II; the physical slot was repurposed in SPEAR3 as a multi-input permit gate and is integral to SPEAR3 machine protection integration.

Conclusion: VXI Slot 5 is **essential** to the SPEAR3 RF protection architecture. It is the **exclusive** integration point for the SPEAR MPS beam permit and orbit interlock into RF station protection. For the RF MPS PLC, it provides redundant defense-in-depth RF cut alongside Path A. Removal of any of its three permit inputs results in: (1) immediate hardware RF drive cut via VXI backplane, and (2) orderly HVPS shutdown via SNL `s_go_off` ~6 s later (but in practice HVPS is hardware-killed within ~20–30 ms via collector overpower cascade — see §5). For the SPEAR MPS and orbit paths, Slot 5 itself has no connection to the SCR ENABLE or crowbar fibers; the hardware HVPS kill is produced by the cascade described in §5.

---

5 Cascade Physics: Secondary Trips Following RF Drive Cutoff

Sources: [R13], [R16], [R23].

When RF drive is cut by a SPEAR MPS permit withdrawal or orbit interlock trip (via Slot 5, or by any mechanism that sets the RFP drive DAC to zero), the immediate RF cutoff creates a secondary physical condition that fires the RF MPS PLC collector overpower protection within one PLC scan cycle. This cascade produces a hardware-speed HVPS kill within ~20–30 ms of the initial RF cut — far faster than the ~6 s SNL orderly shutdown path.

5.1 Mechanism 1 — Klystron Collector Overpower (High Certainty)

A klystron is a velocity-modulation amplifier: it converts cathode beam power into RF output power. When RF drive drops to zero, the klystron immediately stops amplifying — the bunched electron beam produces no RF output. However, as long as the HVPS remains energized, the cathode beam current continues to flow through the klystron body and is deposited in the collector. All cathode input power therefore becomes collector heat dissipation:

$$P_{\text{collector}} = P_{\text{cathode}} - P_{\text{RF out}} \approx P_{\text{cathode}} \quad (\text{when RF drive} = 0,\ \text{HVPS on})$$

During normal CW operation, the klystron operates at some efficiency $\eta = P_{\text{RF out}}/P_{\text{cathode}}$, so the collector absorbs only the fraction $(1-\eta)$ of cathode power. When drive is cut, $P_{\text{RF out}} \to 0$ and the collector must suddenly absorb 100% of cathode power — a step increase proportional to the operating efficiency. For a klystron running at 50% efficiency, this is a $2\times$ step in collector power dissipation.

The RF MPS PLC monitors collector power continuously via a dedicated AB analog input channel:

EPICS PV	Record	Source	Description
`{STN}:KLYSCOLLPLC:POWER`	`ai`	AB Remote I/O T4[41], EGUF=1200 kW	Real-time PLC collector power reading (HIHI=1175 kW MAJOR, HIGH=1150 kW MINOR)

EPICS PV	Record	Source	Description
`{STN}:KLYSCOLL:POWER:ULIM`	`ai`	AB Remote I/O T4[42], EGUF=1200 kW	Configurable PLC trip limit register (operator-set; the PLC drops its collector summary when measured power exceeds this value)
`{STN}:KLYSCOLL:POWER`	`sub` (subIQpowerNet)	INPA=`HVPS:POWER`, INPB=`KLYSOUTFRWD:POWER`	EPICS-computed collector power = cathode power − klystron RF output
`{STN}:KLYSCOLL:POWER:LTCH`	`bi`	AB Remote I/O WL=16, WF=31, B=2	Hardware latch bit set by PLC when it trips on collector overpower

Important distinction: `KLYSCOLLPLC:POWER.HIHI = 1175 kW` is an **EPICS software alarm** that propagates through Channel Access; it does not directly change the PLC outputs. The PLC's own internal comparator — comparing the measured analog value against the limit stored in T4[42] — drops the collector summary to the RF Local Display Panel independently of EPICS. The PLC acts first (~10 ms); the EPICS alarm propagates ~100 ms later via Remote I/O polling.

Cascade signal chain (Figure 3):

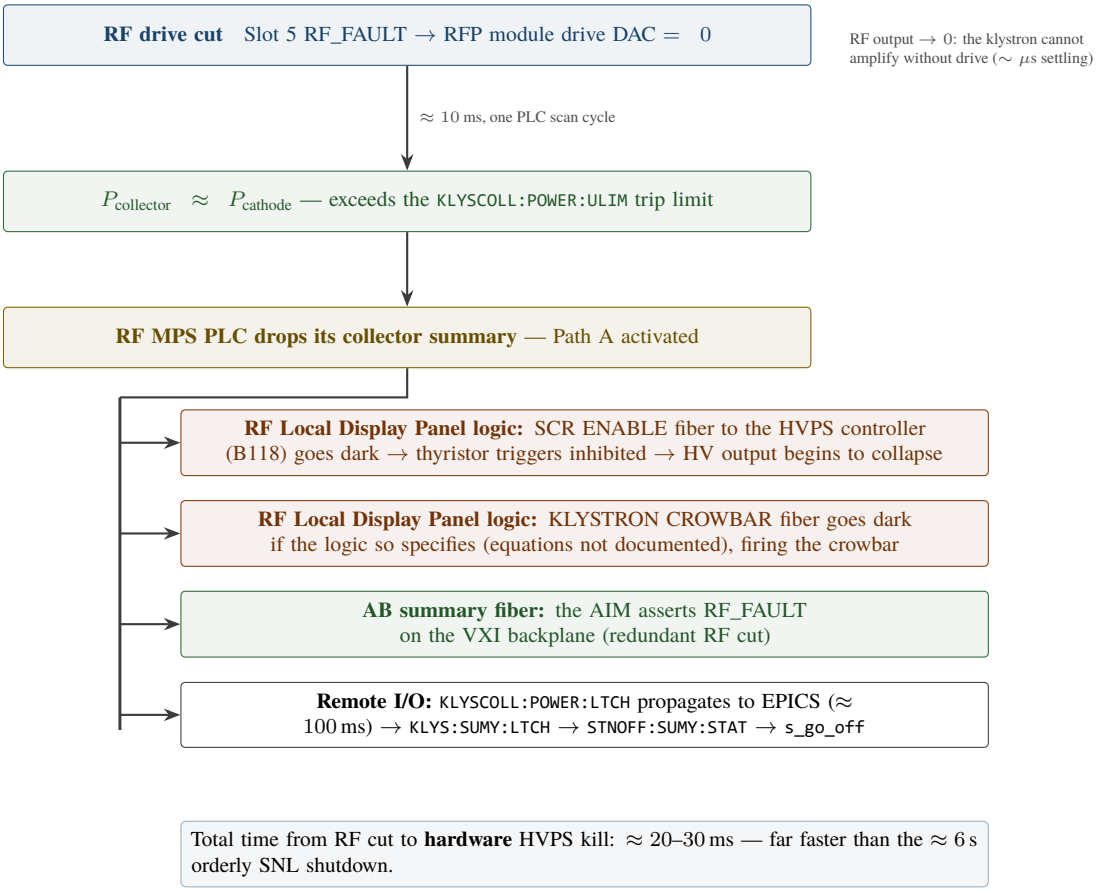


Figure 3: Cascade following an RF drive cut. Secondary trips follow the primary event and can reach the HVPS faster than the SNL shutdown does.

5.2 Mechanism 2 — Reflected Power Transient (Conditional)

When klystron output power drops abruptly, the sudden impedance change at the output waveguide may produce a brief transient in reflected power at the klystron output junction. Operational calibration data records `{STN}:KLYSOUTREFL:POWER` (IQA1 Ch2) HIHI = 10.8 W, HIGH = 8.1 W — these are the EPICS IQA-based software alarm thresholds.

The Fast Interlock Chassis independently monitors reflected power at multiple waveguide points via dedicated analog diode detectors and hardwired comparators. If a transient at any monitored junction exceeds the Fast IC's threshold, the Fast IC trips in < 1 μs and the AIM cuts RF.

However, this mechanism is conditional and less certain than Mechanism 1:

- **Circulator isolation**: The circulator between the klystron output and the cavities is specifically designed to route reflected power (including cavity ring-down) to a load port, providing ~20 dB or better isolation to the klystron. A well-functioning circulator absorbs the transient.
- **Threshold uncertainty**: The Fast IC's analog comparator threshold is not directly software-readable; it may differ from the EPICS IQA HIHI thresholds.
- **Drive-level dependence**: If the klystron was operating at reduced power before the trip, the reflected power transient may be too small to exceed the Fast IC threshold.

Conclusion: Mechanism 1 (collector overpower) is the dominant, reliable secondary trip that fires after any RF drive cutoff with HVPS energized at normal operating power. Mechanism 2 (reflected power transient) is a possible additional trigger but is not guaranteed.

5.3 Net Effect on HVPS Kill Timing

Scenario	HVPS kill mechanism	Time from SPEAR MPS / orbit trip
Software path only (no cascade)	SNL `s_go_off` Step 4: `hvpstrig=OFF`	~6 s
Via collector overpower cascade (normal operation)	RF MPS PLC collector summary → RF Local Display Panel logic: SCR ENABLE removed (crowbar as the logic specifies)	**~20–30 ms**
Via reflected power transient (conditional)	Fast IC analog comparator trip, then the AIM and the panel logic (O4)	< 1 μs to the RF cut (if it fires)

Fault record implication: The cascade trigger appears in the fault record as `{STN}:KLYSCOLL:POWER:LTCH` (collector overpower), not as `STN:MPS:LTCH` or `STN:VXI:LTCH`. The root cause (SPEAR MPS or orbit trip) is visible in those latter PVs but with a ~15–20 ms earlier timestamp. EPICS archiver resolution (~1 Hz default) may not distinguish these — event-driven archiver timestamps are required for precise sequencing.

---

6 RF MPS PLC (PLC-5/1771)

Sources: [R13], [R23].

6.1 Platform

Attribute	Detail
Hardware	Allen-Bradley PLC-5 with 1771 I/O (upgraded from PLC-5 / 1771-DCM)
Location	B132 (same rack as VXI crate)
Remote I/O adapter	**Adapter 3** (`STNDCM`)
Status	Hardware replaced; software written and tested without RF power

6.2 Inputs

Input	Source	Connection	Notes
Klystron collector power (calculated)	Klystron high-voltage and RF output detectors	Analog I/O module in the PLC-5 1771 rack	Cathode power minus RF output; excessive difference = klystron overheating
Reflected power (4 cavities)	RF detector diodes at cavity and waveguide junctions	Analog I/O	VSWR or arc condition in waveguide
Secondary arc sensors	Arc sensor inputs not in Fast IC path	Digital I/O	Redundant arc coverage in waveguide runs
Cooling water flow	Flow meters on klystron and dummy load circuits	Digital/analog I/O	Below minimum flow trips MPS
Cooling water temperature	Temperature sensors on water circuits	Analog I/O	Exceeding maximum trips MPS
Klystron vacuum	Ion pump current or Penning gauge	Analog I/O	Vacuum excursion protection
HVPS fault status	SLC-500 HVPS PLC	Remote I/O (the VXI scanner polls both adapters independently)	HVPS faults forwarded to MPS summary
476 MHz reference power	`{STN}:STNREF:POWER:LTCH`	EPICS gateway	Reference RF monitor
RF Local Display Panel key switches	RF Local Display Panel 340-310 (J2)	24 V lines through cross-connect X530	Local/remote and online/offline modes [R29], [R30]

6.3 Outputs

Output	Destination	Connection	Notes
Collector and waveguide pressure summaries (Path A)	RF Local Display Panel logic (J3)	24 V lines through cross-connect X530	Dropping either removes SCR ENABLE (and fires the crowbar as the panel logic specifies, O4) [R27], [R29]
AB summary and AB heartbeat (Path A)	AIM, through the RF Local Display Panel fibers and the Fast IC	24 V lines (J3), then fiber	AB summary fault → AIM → `RF_FAULT` [R26]
15 area summaries	RF Local Display Panel display (J3)	24 V lines through X530	Status indication only
MPS permit (Path C)	VXI Slot 5 MPS Shutoff module (back connector)	Hardwired contact	De-energizing removes one of Slot 5's three permits → RFP drive cut via VXI backplane
Remote I/O permit status bit	EPICS IOC (via AB-6008 scanner)	RIO word T6[WL,B]	Software path (Path B) — updates `{STN}:STN:MPS:LTCH`

6.4 Protection Functions

The MPS PLC monitors **equipment protection** parameters on a ~10 ms scan cycle:

Monitored Parameter	Trip Condition	Notes
Klystron collector power	Cathode power − RF output exceeds limit	Excessive collector power = poor klystron efficiency → overheating
Reflected power	Reflected power at any cavity exceeds limit	VSWR/arc condition in waveguide
Arc conditions	Secondary arc sensors (not in Fast IC path)	Redundant arc coverage
Cooling water flow	Below minimum flow rate	Klystron and waveguide loads
Cooling water temperature	Exceeds maximum	Thermal runaway detection
Klystron vacuum	Ion pump current or Penning gauge exceeds limit	Vacuum excursion protection
HVPS fault conditions	Status from SLC-500 via Remote I/O	Forwarded to MPS summary
Reference power level	`{STN}:STNREF:POWER:LTCH`	476 MHz reference monitor

6.5 Trip Actions — Three Paths

Path A — Hardware HVPS shutdown + RF cut (fast, ~10 ms PLC scan):

The MPS PLC's hardwired outputs run as 24 V lines through cross-connect X530 to the RF Local Display Panel [R26], [R30]; whether any other hardwired line reaches the Fast Interlock Chassis has not been traced. [O2] When the PLC detects a fault requiring immediate action:

1. The PLC drops the affected summaries and its AB summary
2. The collector and waveguide pressure summaries are inputs of the RF Local Display Panel logic, which removes SCR ENABLE on the fiber to the HVPS controller in B118 and, as its equations specify, fires the crowbar [R27], [R29]
3. The AB summary reaches the AIM over the panel's fiber and the Fast Interlock Chassis → AIM asserts `RF_FAULT` on VXI backplane → RFP RF drive zeroed

This path: ~10 ms (PLC scan) + panel logic and fiber response (μs scale) ≈ **~10–15 ms total**. **Action**: HVPS shutdown (SCR ENABLE) for the faults wired into the panel logic, AND RF drive cut. Other MPS faults, for example cooling or vacuum [R27], remove high voltage only through the collector-power cascade (§5) or the SNL.

Path B — Software/EPICS notification (slow, ~1 s):

1. PLC writes "no permit" status to its Remote I/O output data table
2. VXI AB6008 scanner reads Remote I/O registers at ~1 Hz
3. Updates `{STN}:STN:MPS:LTCH` EPICS record
4. Alarm propagates via `rf_sumy_stn_spr.db` → `{STN}:STNMPS:SUMY:LTCH.INPI` → `{STN}:STNOFF:SUMY:STAT.SEVR`
5. SNL `fault_stnoff != NO_ALARM` triggers `s_go_off`

Path B is **bookkeeping** — the MPS hardware has already acted on Path A. Path B ensures the EPICS state machine learns about the fault, logs it, triggers fault file capture, and performs the orderly shutdown sequence. **Action**: Orderly HVPS ramp-to-zero + fault file capture (hardware already tripped via Path A).

Path C — VXI backplane RF permit gate (~10 ms PLC scan):

The same PLC fault that triggers Path A also removes the RF MPS PLC permit from the VXI Slot 5 MPS Shutoff module (back connector). The module asserts `RF_FAULT` (or removes `RF_PERMIT`) on the VXI P2 backplane → RFP RF drive cut.

Path C: ~10 ms (PLC scan) + output response ≈ **~10 ms total**. **Action**: Immediate RF drive cut. No immediate HVPS hardware action. HVPS orderly shutdown follows via SNL `s_go_off` (~6 s).

Note on SPEAR MPS and orbit interlock: These two signals are wired **directly** to the Slot 5 back connector — they enter Slot 5 independently of the RF MPS PLC. Their removal triggers `RF_FAULT` on the VXI backplane (same immediate RF drive cut as Path C above) but does **not** trigger Path A, so there is no immediate HVPS hardware shutdown. However, the `RF_FAULT` line asserted by Slot 5 is seen by the AIM module's ISR, which propagates through the EPICS alarm chain (`STN:VXI:LTCH` → `STNPARK:SUMY:STAT` → `STNOFF:SUMY:STAT` → `fault_stnoff`) and causes the SNL `s_go_off` to execute an orderly HVPS shutdown ~6 s later. The RF MPS PLC does not relay these signals to the Fast IC or the RF Local Display Panel. See §4.2 for the complete Slot 5 input list and §4.5 for the alarm chain analysis.

Note: Path C provides a redundant VXI-backplane RF cut that is independent of the AIM module's health. The immediate HVPS shutdown is owned by Path A through the RF Local Display Panel logic.

6.6 EPICS PV Path

Figure 4 summarizes the PLC's inputs and outputs.

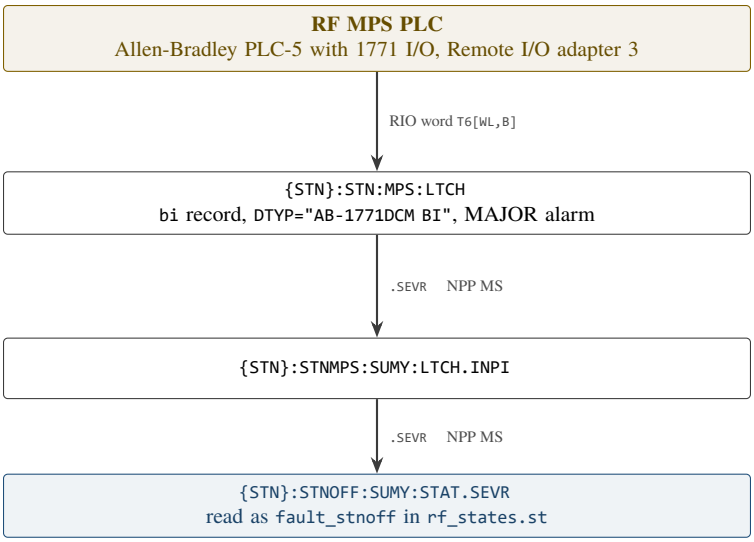


Figure 4: RF MPS PLC (Allen-Bradley PLC-5 with 1771 I/O) inputs and outputs.

6.7 MPS Wiring Documents

33 MPS wiring diagrams (SLAC drawings wd3403300200 through wd3403303400) describe the complete MPS hardware signal chain. Original harness is for the PLC-5.

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7 SLC-500 HVPS PLC

Sources: [R4], [R8], [R9], [R10], [R15], [R16], [R23], [R24], [R25].

7.1 Platform

Attribute	Detail
CPU	Allen-Bradley 1747-L532
Remote I/O adapter module	Allen-Bradley 1747-DCM, chassis Slot 1, full rack
Location	B118 (HVPS building), Hoffman Box enclosure
Remote I/O adapter	**Adapter 1** (HVPSDCM, full rack)
Rack	10-slot SLC chassis (slots 0–9; slot 4 empty)

7.2 Inputs

Input	Source	Connection	Notes
HVPS DC output voltage	HVPS power supply analog	Slot 8–9 Analog I/O (AI word 30)	Used for voltage readback and overvoltage detection

Input	Source	Connection	Notes
HVPS DC output current	HVPS power supply analog	Slot 8–9 Analog I/O (AI word 31)	Used for current readback and overcurrent detection
Primary AC current	AC current transformer	Slot 8–9 Analog I/O	Overcurrent detection (`HVPSAC:CURR:LTCH`)
Oil temperature (thermocouple)	Oil tank thermocouple	Slot 3 thermocouple input	`HVPSOIL:TEMP:LTCH` overtemperature latch
PPS enable	PPS system (GOB1208PNE, Pin E→F)	Slot 6 IB16 digital input 14	Used for HV vacuum contactor control
SCR ENABLE and KLYSTRON CROWBAR fibers	**RF Local Display Panel (B132)**	**HFBR-2412 receivers on the right-side and left-side trigger interconnect boards**	**The boards act on the fibers directly: a dark SCR ENABLE inhibits the thyristor triggers and a dark KLYSTRON CROWBAR fires the crowbar [R27], [R28]. The boards' monitoring bus carries their state (SCR ENABLE MON and others); its PLC input assignment is part of O5. The HVPS STATUS fiber is an output of the left-side board (§7.3).**
Crowbar fired / arc status	Internal HVPS self-monitoring	Slot 6 IB16 digital inputs	`HVPS:CROWBAR:LTCH`, `HVPSKLYS:ARC:LTCH`, etc.
Oil level	Float switch in HV tank	Slot 6 IB16 digital input	`HVPSOIL:LEVEL:LTCH`
Transformer arc	Arc detector inside HV transformer	Slot 6 IB16 digital input	`HVPSXFORM:ARC:LTCH`
SCR bank status	SCR gate driver feedback	Slot 6 IB16 digital inputs	`HVPSSCR1:ON:STAT`, `HVPSSCR2:ON:STAT`
Contactor status	HV vacuum contactor auxiliary contacts	**Slot 7 IV16** digital inputs 0–3	Blocking relay (IN0), overcurrent relay (IN1), contactor closed (IN2), contactor ready (IN3) — `HVPSCONTACT:*:STAT` group. These sit on the **IV16 in slot 7**, not on the slot-6 IB16
`hvpstrig` (SCR enable command)	EPICS IOC via Remote I/O (SNL `HVPSONSUB`)	Remote I/O incoming data table → Slot 5 OX8 output 0	Supervisory SCR enable gate; must be ON for HVPS to produce HV

7.3 Outputs

Output	Destination	Connection	Notes
24 V fault/enable (`HVPSSCR:ON:CTRL`)	Trigger interconnect boards (COMMANDS bus), same B118 enclosure	Slot 5 OX8 output 0; no fiber [R28]	Supervisory thyristor-trigger enable; ~20 ms speed; must be on together with the SCR ENABLE fiber from the RF Local Display Panel
HVPS STATUS fiber	RF Local Display Panel (B132), received as the HVPS beam abort	HFBR-1412 transmitter on the left-side trigger interconnect board	Lit while the 24 V fault/enable is high and no crowbar is commanded; it says nothing about whether the thyristors fire [R27], [R28]

Output	Destination	Connection	Notes
Ross grounding switch command	Ross Engineering HQ3 grounding switch	Slot 2 IO8 (120 VAC)	**PPS Chain 2**: grounds HV tank before personnel entry
HV vacuum contactor command	Ross Engineering HQ3 vacuum contactor	Slot 5 OX8 OUT2 → relay chain (K4 → MX → RR → L1)	**PPS Chain 1**: connects/disconnects 12.47 kV AC to HVPS primary (fail-safe open)
HVPS Remote I/O status registers	EPICS IOC (via AB-6008 scanner, adapter 1)	Remote I/O	All `HVPS*:*:LTCH` and `HVPS*:*:STAT` PVs populated from these registers
Voltage setpoint output	HVPS voltage control SCR firing angle	Slot 8–9 Analog I/O (AO)	SNL `rf_hvps_loop.st,v` writes ramp-controlled setpoint

7.4 Rack Slot Configuration

Slot	Module	Function
0	1747-L532 CPU	Program execution
1	1747-DCM	Remote I/O adapter module — presents the SLC-500 chassis to the VXI scanner as adapter 1 (full rack)
2	IO8 (8-pt, 120 VAC)	Ross grounding switch output (PPS Chain 2)
3	Thermocouple input	HVPS temperature measurements
5	OX8 relay output	Output 0: 24 V fault/enable to the trigger boards (see §7.6); output 2: contactor relay chain
6	IB16 digital input	PPS enables, HVPS status inputs
8–9	Analog I/O	HVPS voltage readback, current readback, setpoint output

7.5 What the SLC-500 Interlocks On

The SLC-500 monitors every significant HVPS internal condition and exposes them as EPICS records via Remote I/O. The full set of HVPS interlock sources, with their Remote I/O word/bit assignments derived from `rf_digital_All.substitutions,v` [R9] — these have not been cross-checked against the deployed ladder program [O5]:

7.5.1 Category 1 — Electrical Faults

PV	RIO Card/Bit	Alarm	Description
`{STN}:HVPS:CROWBAR:LTCH`	C12.9	MAJOR	**Crowbar fired** — HVPS crowbar thyristor discharged; stored energy dumped

PV	RIO Card/Bit	Alarm	Description
`{STN}:HVPS:VOLT:LTCH`	C14.1	MAJOR	**Overvoltage** — DC output exceeded rated voltage
`{STN}:HVPSAC:CURR:LTCH`	C12.3	MAJOR	**AC current trip** — primary AC current exceeded limit
`{STN}:HVPS:OPENLOAD:LTCH`	C14.9	MAJOR	**Open load** — no current path to load (klystron disconnected or contactor open during energization)
`{STN}:HVPS:PANIC:LTCH`	C12.11	MAJOR	**Emergency Off (PANIC)** — manual or automatic emergency shutdown latch
`{STN}:RFHVPS:CROWBAR:LTCH`	C14.11	MAJOR	**RF crowbar request** — request from RF system to fire HVPS crowbar

7.5.2 Category 2 — Oil System

PV	RIO Card/Bit	Alarm	Description
`{STN}:HVPSOIL:LEVEL:LTCH`	C12.14	MAJOR	**Oil level low** — insulating oil in HV tank below minimum
`{STN}:HVPSOIL:TEMP:LTCH`	C12.15	MAJOR	**Oil temperature high** — insulating oil overtemperature
`{STN}:HVPSOIL:TEMP`	Analog (SLC-500 AI)	Monitored	Oil temperature value (°C); also has configurable ULIM at `{STN}:HVPSOIL:TEMP:ULIM`

Design note: HVPS oil temperature was removed from the `HVPSSTN:SUMY:LTCH` roll-up in the Sass 2004 revision (`rf_sumy_hvps.db,v` rev 1.2 log: "Remove HVPS oil temperature from HVPS summary. It remains in the overall temperature summary"). Oil temperature LTCH still contributes to `{STN}:HVPSTEMP:SUMY:LTCH` → `{STN}:STNTEMP:SUMY:LTCH` path.

7.5.3 Category 3 — Transformer / HV Tank

PV	RIO Card/Bit	Alarm	Description
`{STN}:HVPSXFORM:ARC:LTCH`	C14.10	MAJOR	**Transformer arc** — arc detector inside the HV transformer tank
`{STN}:HVPSXFORM:PRESS:LTCH`	C14.6	MAJOR	**Transformer overpressure** — SF or oil pressure in transformer enclosure exceeded limit
`{STN}:HVPSXFORM:VACM:LTCH`	C14.7	MAJOR	**Transformer vacuum/pressure** — vacuum integrity or gas pressure fault
`{STN}:HVPSKLYS:ARC:LTCH`	C12.4	MAJOR	**Klystron arc** (as seen from HVPS side) — arc current surge into cathode load
`{STN}:HVPS:TEMP:LTCH`	C14.0	MAJOR	**Overtemperature** — general HVPS enclosure over-temperature

7.5.4 Category 4 — Supply and Contactor Status

PV	RIO Card/Bit	Alarm	Description
{STN}:HVPS:PPS:STAT	C14.8	MAJOR on 0=NOPERMIT	**PPS Status** — PPS permit present (PERMIT=0, NOPERMIT=1 → alarm inverted)
{STN}:HVPS12KV:VOLT:STAT	C12.1	MAJOR on 0=NOTAVAIL	**12 kV AC available** — primary 12.47 kV supply status
{STN}:HVPSAC:POWER:STAT	C12.2	MAJOR on 0=OFF	**Auxiliary AC power** status
{STN}:HVPSDC:POWER:STAT	C12.10	MAJOR on 0=OFF	**Auxiliary DC power** status
{STN}:HVPSSUPPLY:ON:STAT	C14.4	MAJOR on 0=OFF	**HV supply on** status
{STN}:HVPSSUPPLY:READY:STAT	C14.5	MAJOR on 0=NOTREADY	**HV supply ready** status
{STN}:HVPSENERFAST:ON:STAT	C12.12	MAJOR on 0=OFF	**Fast inhibit** — energization fast inhibit active
{STN}:HVPSENERSLOW:START:STAT	C12.13	MAJOR on 1=INACTIVE	**Slow start** — energization slow-start sequencer
{STN}:HVPSCONTACT:CLOSE:STAT	C12.5	MAJOR on 0=OPEN	HV contactor close status
{STN}:HVPSCONTACT:ON:STAT	C12.6	MAJOR on 0=OFF	HV contactor on status
{STN}:HVPSCONTACT:OPEN:STAT	C12.7	MAJOR on 1=OPEN	HV contactor open status
{STN}:HVPSCONTACT:READY:STAT	C12.8	MAJOR on 0=NOTREADY	HV contactor ready
{STN}:HVPSSCR1:ON:STAT	C14.2	MAJOR on 0=OFF	SCR bank 1 on status
{STN}:HVPSSCR2:ON:STAT	C14.3	MAJOR on 0=OFF	SCR bank 2 on status

Note on Remote I/O addressing: ”C12.9” means Card (output word) 12, bit 9. Word 12 = physical I/O module card 12÷2 = slot 6. The SLC-500 uses 8-bit word addressing where adjacent card pairs share a 16-bit word.

7.5.5 Category 5 — Analog Monitoring

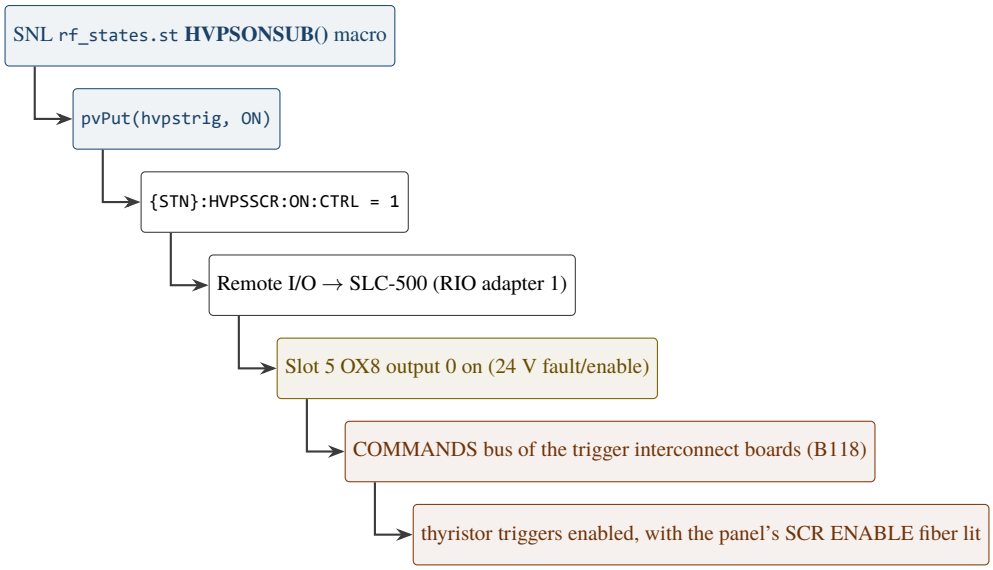


Figure 5: HVPS enable chain executed by the `HVPSONSUB()` macro in `rf_states.st`.

PV	Source	Alarm Limits	Description
`{STN}:HVPS:VOLT`	SLC-500 AI	HIHI=87, HIGH=85 kV	HVPS DC output voltage
`{STN}:HVPS:CURR`	SLC-500 AI	HIHI=30, HIGH=30 A	HVPS DC output current
`{STN}:HVPS:PLC:VOLT`	SLC-500 AI word 30	±100 kV range	Voltage readback from PLC (independent ADC)
`{STN}:HVPS:PLC:CURR`	SLC-500 AI word 31	±50 A range	Current readback from PLC (independent ADC)
`{STN}:HVPSAC:CURR`	SLC-500 AI	Monitored	Primary AC current
`{STN}:HVPSOIL:TEMP`	Thermocouple slot 3	ULIM configurable	Oil temperature (°C)

7.6 SCR Enable Output — The HVPS Supervisory Enable Path

The SLC-500 enables the thyristor triggers through output 0 of its slot-5 relay module (24 V fault/enable), which drives the COMMANDS bus of the trigger interconnect boards in the same B118 enclosure; no fiber is involved [R28]. This is the **supervisory SCR ENABLE path** — it must be held on for the HVPS to maintain output voltage, and the HVPS STATUS fiber is lit only while it is on. Figure 5 shows the sequence that establishes it.

This path is **distinct from** the SCR ENABLE fiber formed by the RF Local Display Panel:

Path	Speed	Source	Notes
RF Local Display Panel → SCR ENABLE (fiber)	μs	Panel logic	Hardware permit, removed on fault

Path	Speed	Source	Notes
SLC-500 output 0 → COMMANDS bus (24 V)	~20 ms	Supervisory enable/disable	Must be explicitly enabled; cleared in s_go_off

Both act on the trigger interconnect boards in B118: the panel's path is a fiber from B132, the SLC-500's is wired on the COMMANDS bus. **Both** must be asserted for the thyristors to fire. The panel path is the fast-trip path; the SLC-500 path is the supervisory gate that must be deliberately enabled by the SNL state machine as part of the startup sequence.

PV: `{STN}:HVPSSCR:ON:CTRL` (bo record, `rf_hvps.db,v`): "HVPS SCR Enable Switch". Writing ON enables the HVPS; writing OFF disables (part of s_go_off orderly shutdown).

7.7 HVPS Alarm Aggregation Tree

Figure 6 shows how the individual HVPS latches roll up into the three summary records.

7.8 PPS Dual-Chain Architecture (Contactor and Grounding Switch)

	Chain 1 — HV Vacuum Contactor	**Chain 2 — Ross Grounding Switch**
Safety function	Disconnect 12.47 kV AC primary power from HVPS	Ground the HVPS HV bus (−77 kV) for safe personnel access
SLC-500 output	Slot-5 OX8 OUT2 relay contact → K4 → MX → L1 relay chain	Slot-2 IO8 OUT3 (120 VAC) → Ross switch coil
PLC rung	Rung 0017 (mislabeled "Crowbar On" on original drawing — corrected to "Contactor Enable")	Rung 0016 (Ross switch enable)
PPS input	PPS 1 (GOB1208PNE Pin E→F)	PPS 2 (GOB1208PNE Pin G→H)
Operating state	Coil held energized (contactor **CLOSED** — 12.47 kV connected)	Coil held energized (switch held **OPEN** — HV bus not grounded)
Fail-safe state	PPS removed → K4 drops → L1 drops → contactor **spring-opens** → 12.47 kV disconnected	PPS removed → 120 VAC removed → switch **spring-closes** → HV bus grounded
Readback signal	S5 NC auxiliary contact → GOB1208PNE Pins A-B	Ross switch NC auxiliary contact → GOB1208PNE Pins C-D
Readback meaning	Pins A-B shorted = contactor OPEN (12.47 kV removed, safe)	Pins C-D shorted = switch CLOSED (HV bus grounded, safe)

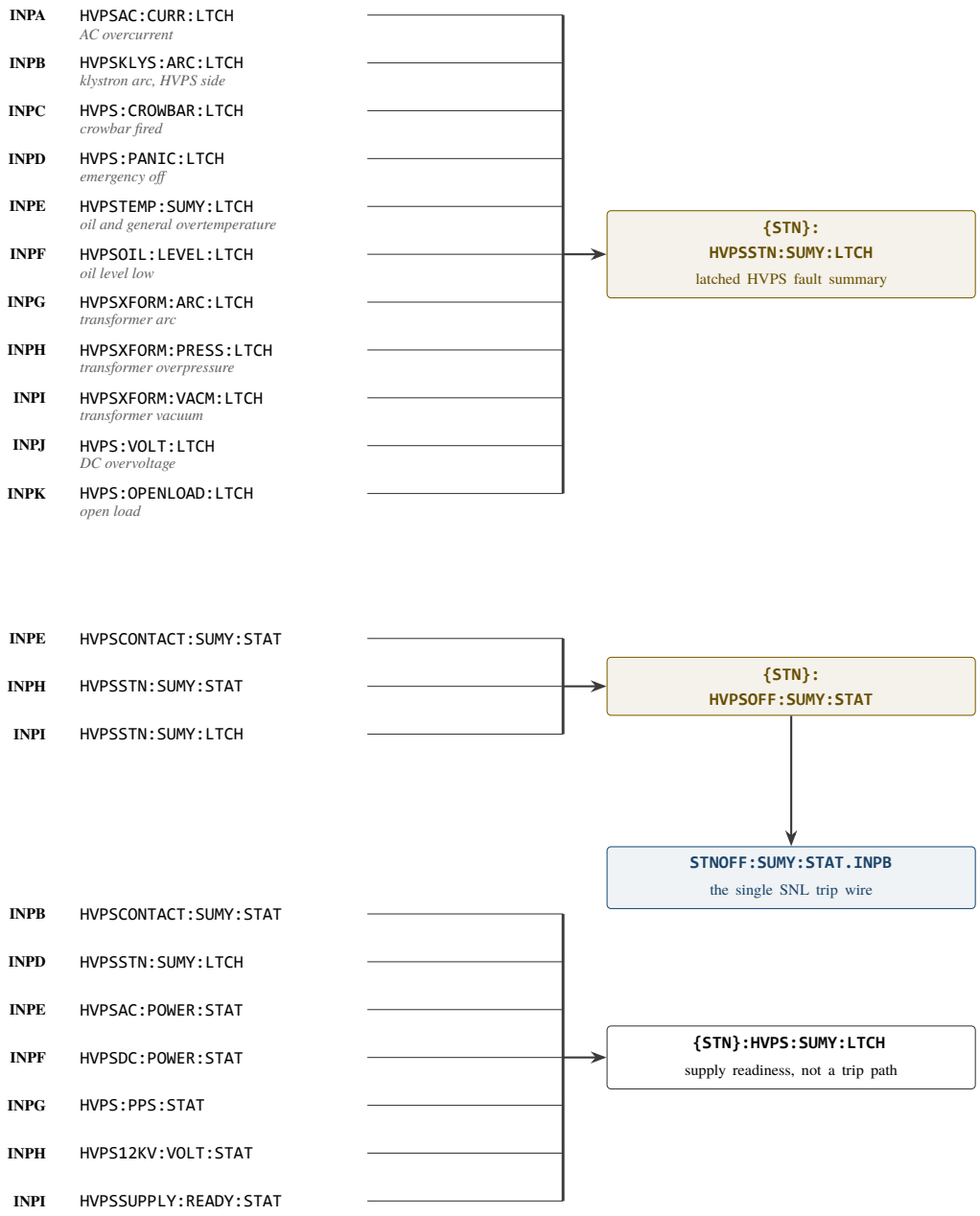


Figure 6: HVPS alarm aggregation. Three summary records, each collecting its inputs on a spine. `HVPSSTN:SUMY:LTCH` is itself an input to both of the others (as INPI and INPD respectively), so a single latched HVPS fault reaches the station trip wire through `HVPSOFF:SUMY:STAT`. `HVPS:SUMY:LTCH` is a readiness summary and is *not* a trip path.

7.8.1 Chain 1 — HV Vacuum Contactor Relay Path

The full relay chain from PPS to contactor, as routed through the SLC-500. This is the personnel-safety path described in `tex/docL-legacy-architecture/L_legacy_system_architecture.pdf` (PPS interface).

```
PPS Enable (GOB1208PNE, Pin E→F)
    → SLC-500 Slot-6 IB16 Input 14
    → PLC Rung 0017 (ladder logic)
    → Slot-5 OX8 OUT2
    → Terminal Strip TS-5 → switchgear cable
    → K4 relay (PPS control)
    → MX relay (external operational enable)
    → RR relay (reset latch)
    → L1 holding coil
    → Ross Engineering HQ3 vacuum contactor
      (connects/disconnects 12.47 kV AC to HVPS primary)
```

7.8.2 Chain 2 — Ross Grounding Switch

```
PPS 2 Enable (GOB1208PNE Pin G→H)
    → SLC-500 PLC Slot-6 IB16 Input 15
    → PLC Rung 0016 (Ross switch enable)
    → Slot-2 IO8 OUT3 (120 VAC)
    → TS-6 → Belden 83709 cable → Termination Tank
    → Ross Grounding Switch coil

Readback: Ross Switch NC Aux → TS-6 pins 11,12 → GOB1208PNE Readback C-D
```

Critical design issue: The PPS chain routes through SLC-500 ladder logic — a programmable device is in the personnel safety chain. This does not meet modern PPS standards (SLAC ES&H). Flagged in `tex/docL-legacy-architecture/L_legacy_system_architecture.pdf` (identified PPS issues).

7.8.3 Fail-Safe Directions

Both chains are designed so that removing electrical power drives them to the personnel-safe state:

- **Chain 1** (contactor): Operator/PPS removes PPS 1 → K4 relay de-energizes → holding coil L1 loses drive → vacuum contactor spring-opens → 12.47 kV AC is removed from the HVPS primary transformer. A power loss anywhere in the relay chain produces the same outcome.
- **Chain 2** (grounding switch): Operator/PPS removes PPS 2 → PLC Rung 0016 no longer energizes 120 VAC output → Ross switch coil de-energizes → switch spring-closes → HVPS HV bus is grounded. A power loss to the PLC or 120 VAC supply produces the same outcome.

Both readbacks use **NC (normally-closed) auxiliary contacts**, meaning the readback circuit is closed (complete) only when the device is at its safe-access state. A broken wire or failed contact reads as ”unsafe” — fail-safe readback.

7.8.4 Safe-Access Operational Sequence

Personnel access to the SSRL tunnel requires both chains in their safe state **in the correct order**:

```
SHUTDOWN for personnel entry:
  1. PPS 1 removed → Rung 0017 drops → contactor opens → 12.47 kV disconnected
  2. HVPS energy dissipates (capacitor discharge through load)
  3. PPS 2 removed → Rung 0016 drops → Ross switch closes → HV bus grounded
  4. Readbacks A-B and C-D loop complete → PPS confirms both chains safe
```

```
  5. Personnel entry permitted

RESTORE after personnel exit:
  1. Personnel withdraw; PPS confirms exclusion at entry points
  2. PPS 2 restored → Rung 0016 energized → Ross switch opens (HV bus un-grounded)
  3. PPS 1 restored → Rung 0017 energized → K4 → contactor closes → 12.47 kV restored
  4. Normal HVPS startup sequence may proceed
```

Critical: The contactor (Chain 1) must be OPEN and HVPS fully discharged **before** the grounding switch (Chain 2) closes onto the HV bus. Closing a grounding switch onto a live high-voltage bus would produce a catastrophic high-energy arc fault. The site PPS system enforces correct sequencing at the machine level through PPS permitting logic.

7.8.5 Compliance Comparison

	Chain 1	Chain 2
Compliance issue	PPS 1 routes through SLC-500 Rung 0017	PPS 2 routes through SLC-500 Rung 0016
Hardware fail-safe	**Present** — OX8 relay input side is wired directly to PPS 1 signal (24VDC from GOB1208PNE Pin E). Even if the PLC closes the output contact, K4 coil cannot energize without PPS 1 voltage physically present.	**Absent** — PLC drives 120 VAC directly to Ross switch coil via IO8 OUT3. If PLC fails energized (maintains 120 VAC output despite PPS 2 removal), the Ross switch remains held open (un-grounded) without a PPS 2 command.
Failure mode concern	PLC closes OX8 spuriously → K4 cannot energize (no PPS 1 voltage) → **no hazard**	PLC maintains 120 VAC spuriously → Ross switch stays open → **HV bus not grounded when PPS removed**

Chain 2 is the higher-priority compliance concern. The spring-return mechanism provides a fail-safe against total power loss, but does not protect against a PLC output failure that maintains the coil energized. The upgrade architecture (see `pps/HoffmanBoxPPSWiring.docx`) eliminates this by providing direct PPS control of both chains through the Interface Chassis, bypassing the PLC for safety-critical permitting.

Sources: [R25] (detailed wiring); `pps/HoffmanBoxPPSWiring.docx`.

---

8 SNL State Machine (`rf_states.st,v`)

Sources: [R2], [R11], [R12], [R22].

8.1 Inputs

Input PV	Source	Connection	Role
{STN}:STNOFF:SUMY:STAT.SEVR (fault_stnoff)	EPICS alarm aggregation tree (rf_sumy_stn.db)	EPICS CA pvMonitor	**Master trip wire** — any MAJOR alarm propagated here triggers s_go_off
{STN}:HVPS:VOLT (hvpsvoltrd)	SLC-500 AI (RIO word 30)	EPICS CA pvGet	HVPS voltage readback for ramp control
{STN}:HVPS:CURR	SLC-500 AI (RIO word 31)	EPICS CA pvGet	HVPS current readback
{STN}:STN:AIM:ARCLTDSTT	AIM ARCLTDSTT register	EPICS CA pvMonitor	Arc latched status — used to distinguish arc fault from other faults
{STN}:STN:FORCED:LTCH (forced_fault)	AIM first-fault register bit	EPICS CA pvMonitor	Hard arc latch — blocks auto-reset
{STN}:HVPSOFF:SUMY:STAT	EPICS HVPS alarm tree	EPICS CA pvMonitor	HVPS off/fault status, feeds fault_stnoff
{STN}:STN:LOCAL:ON	Local panel switch	EPICS CA	Local/remote mode; local mode forces OFF
{STN}:STN:STATE (stnstate)	EPICS state PV	EPICS CA	Station state (restored at boot in s_init)
{STN}:HVPSSCR:ON:CTRL readback	Remote I/O readback from SLC-500	EPICS CA	SCR enable relay state confirmation
TAXI/GVF status (via rf_msgs.st,v)	GVF module software PVs	EPICS inter-IOC	TAXI error monitoring; triggers resync if TAXI errors detected

8.2 Outputs

Output PV	Destination	Action
{STN}:STN:AIM:FRCBMABT (fba)	AIM MODU.FBA → RF Local Display Panel → beam-abort output	Force Beam Abort — station not ready for beam; set in s_go_off Step 1
{STN}:STN:AIM:MODU.RBA (rba)	AIM MODU.RBA	Reset Beam Abort — de-asserts the forced beam abort during recovery
{STN}:STN:AIM:MODU.RSTF (rstf)	AIM hardware latches	Resets all AIM arc latches and fast interlock latches
{STN}:STN:AIM:MODU.HVPS (aimon)	AIM HVPS_On → RF Local Display Panel (HVPS on request)	Hardware gate for SCR ENABLE; set in HVPSONSUB
{STN}:HVPSSCR:ON:CTRL (hvpstrig)	SLC-500 Remote I/O → Slot 5 OX8 output 0 → trigger boards	Supervisory SCR ENABLE gate; ON in HVPSONSUB, OFF in s_go_off Step 5
{STN}:HVPS:VOLT:SP (hvpswdefault)	SLC-500 Remote I/O → Analog output → SCR firing angle	HVPS voltage setpoint; set to 0 in s_go_off Step 3

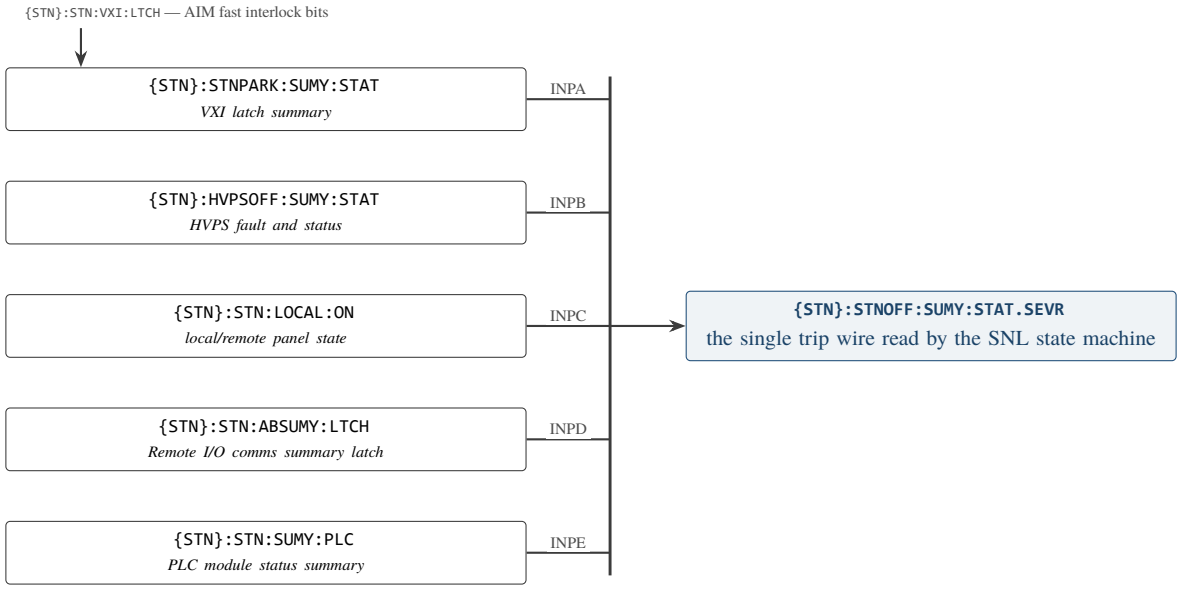


Figure 7: The single SNL trip wire: every fault source converges on `STNOFF:SUMY:STAT.SEVR`.

Output PV	Destination	Action
`{STN}:STN:RFP:RFENABLE` (`rfswitch`)	RFP module RF drive enable	RF drive gate; ON in `HVPSONSUB`, OFF in `s_go_off` Step 6
`{STN}:STN:STATE` (`stnstate`)	EPICS state display PV	Operator state display: OFF/PARK/TUNE/ON_FM/ON_CW
`{STN}:STN:INTCOMP` (`intcomp`)	RFP/IQA integrator compensation	Direct feedback integrator; disabled in `s_go_off` Step 7
`ef_set(ffwrite_ef)`	`ss rf_statesFF` event flag	Triggers fault waveform capture state set in `s_go_off` Step 8

8.3 Architecture

The SNL state machine (`rf_states.st,v`, 2,227 lines, 3 concurrent state sets) is the **software coordination layer**. It does not provide fast protection — by the time `fault_stnoff` changes, all hardware trips have already fired. Its roles are:

1. **Orderly shutdown** — ramp HVPS to zero before removing SCR enable (protects the HV capacitors/diodes from abrupt current interruption)
2. **Fault recording** — capture of 11 signal channel waveforms to circular buffer
3. **Recovery sequencing** — managed restart with `forced_fault` and auto-reset logic
4. **Operator state interface** — OFF/PARK/TUNE/ON_FM/ON_CW state display
5. **BATS management** — force/reset beam abort in the correct sequence

8.4 The Single Trip Wire: `fault_stnoff`

Figure 7 traces every fault source onto the one record the state machine watches.

Any input going MAJOR alarm propagates (via EPICS `MS` = Maximize Severity) to `STNOFF:SUMY:STAT.SEVR`. Every ON state checks this:

```
/* In s_on_cw, s_tune, s_on_fm */
when (fault_stnoff != NO_ALARM) {
    fault_detected = 1;
} state s_go_off
```

8.5 STNMPS Alarm Tree (SPEAR3-specific, `rf_sumy_stn_spr.db`)

The SPEAR3-specific MPS summary aggregates into the station trip path:

```
STN:STNMPS:SUMY:LTCH
    INPB: STN:STN:FORCED:LTCH    (AIM forced fault latch)
    INPC: STN:STN:VXI:LTCH       (AIM fast interlock status bits)
    INPE: STN:STN:NCV:PLC        (no-current-value PLC)
    INPF: STN:STNREF:POWER:LTCH  (476 MHz reference power)
    INPG: STN:STNCRATEPS:TEMP:LTCH (VXI crate power supply temperature)
    INPH: STN:STN:RF476MHZREF:LTCH (476 MHz reference status)
    INPI: STN:STN:MPS:LTCH        (RF MPS PLC permit)
```

8.6 Orderly Shutdown Sequence (`s_go_off`)

```
State s_go_off (entered when fault_stnoff != NO_ALARM in any ON state):

Step 1: TUNESUB()
    fba = 1   pvPut(fba)        → STN:STN:AIM:FRCBMABT = 1
                                   → MODU.FBA asserted → VXI beam abort (beam abort)
    runmode = TUNE               → puts IQA/RFP in tune mode (low drive)

Step 2: setiqtune()              → zero I/Q drive setpoints

Step 3: pvPut(hvpswdefault, 0)  → write 0 to HVPS voltage setpoint
                                   rf_hvps_loop.st ramps voltage down

Step 4: taskDelay(300)           → ~5 second wait for HVPS to discharge

Step 5: hvpstrig = OFF   pvPut(hvpstrig)
    → STN:HVPSSCR:ON:CTRL = 0  → Remote I/O → SLC-500 (RIO adapter 1)
    → SCR gate drive disabled from supervisory path

Step 6: rfswitch = OFF   pvPut(rfswitch)
    → STN:STN:RFP:RFENABLE = 0 → RF drive enable explicitly removed

Step 7: intcomp = OFF    pvPut(intcomp)
    → Direct feedback integrator compensation disabled

Step 8: efSet(ffwrite_ef)        → triggers ss rf_statesFF (fault file capture)

Step 9: → state s_off
```

8.7 HVPS Startup Sequence (HVPSONSUB macro)

```
/* executed inside s_go_on_cw when transitioning to ON_CW */
HVPSONSUB():
    rfswitch = ON          pvPut(rfswitch)   → RF drive enabled
    pvGet(hvpswdefault)                       → read default voltage setpoint
    /* call TUNESUB to home tuners */
    hvpstrig = ON          pvPut(hvpstrig)   → SCR gate drive enabled via Remote I/O
    aimon = ON             pvPut(aimon)       → write {STN}:STN:AIM:MODU.HVPS = 1
```

```
                                                → AIM fiber "HVPS_On" asserted
    /* wait up to 10 s checking fault_stnoff */
```

The aimon write is a **hardware interlock gate**: the AIM module physically holds its "HVPS_On" output LOW until the IOC writes a 1 to MODU.HVPS. The signal reaches the RF Local Display Panel as the HVPS on request, an input of the logic that forms SCR ENABLE [R26], [R27], so the HVPS cannot be energized unless the software state machine has explicitly authorized it.

8.8 BATS Reset (RESET_BMABTSUB)

```
/* RESET_BMABTSUB(fault) --- called during recovery */
if (fault == NO_ALARM) {
    fba = 0;    pvPut(fba)    → de-assert Force Beam Abort
    pvPut(rba)                 → Reset Beam Abort (MODU.RBA)
    pvPut(rstf)                → Reset Faults (MODU.RSTF --- clears all arc latches)
}
```

8.9 forced_fault Latch — Blocking Auto-Reset

{STN}:STN:FORCED:LTCH is set by AIM hardware on detection of a severe arc event. When this latch is set:

1. {STN}:STNMPS:SUMY:LTCH.INPB is MAJOR → contributes to STNOFF:SUMY:STAT
2. In s_go_stn_reset (auto-reset state):

```
   when (forced_fault != NO_ALARM) {
       /* exit without reset attempt */
   } state s_off
```

3. The station stays locked in s_off until an operator physically presses the **FAULT RESET** button on the Fast Interlock Chassis front panel — this clears the AIM hardware latch and de-asserts forced_fault

This was a SPEAR3-specific customization (Sass comment 2004: *"SPEAR is using forced fault to trip the station under certain conditions"*) to prevent automatic recovery after hard arcs.

---

9 Fault Event Timeline

9.1 Example: Waveguide Arc

t = 0 ns Arc breakdown at cavity waveguide window:
→ fiber-optic light pulse arrives at Fast Interlock Chassis

t < 1 μs Fast IC analog comparator fires:
→ Status word updated → AIM module received
→ If the AIM's fault logic drops the HVPS on request or forces
the beam abort, the RF Local Display Panel logic removes SCR ENABLE
(fiber → HVPS controller, B118) and, as its equations specify,
fires the crowbar (O4); HV output then begins to collapse

t < 1 μs AIM hardware asserts RF_FAULT on VXI P2 backplane:
→ RFP module (Slot 4) monitors RF_FAULT pin directly
RF drive DAC output zeroed in hardware
RF power into klystron collapses

t ≈ 5–50 μs AIM ISR fires:
→ Reads ARCLTDSTT latch register (not ISR register — retains fault)
→ forced_fault latch set if arc current exceeded threshold
→ scanOnce({STN}:STN:AIM:MODU)

t ≈ 50–500 μs EPICS IOC processes AIM:MODU record:
→ FANO1/2/3 fanout updates ARCCURSTT, ARCLTDSTT, FSTFLT records
→ {STN}:STN:VXI:LTCH goes MAJOR
→ {STN}:STNMPS:SUMY:LTCH propagates (MS) → STNOFF:SUMY:STAT MAJOR

t ≈ 10–15 ms MPS PLC (PLC-5/1771) next scan cycle:
→ Detects the fault through its own inputs, if any, and sets its
AB summary fault (fiber → AIM); the collector summary follows
if the RF cut raises collector power (§5)
→ Writes STN:MPS:LTCH bit OFF in Remote I/O output data table

t ≈ 20 ms SLC-500 detects HVPS crowbar event, if the crowbar fired:
→ Updates HVPS:CROWBAR:LTCH = FAULT in Remote I/O registers
→ HVPSSTN:SUMY:LTCH → HVPSOFF:SUMY:STAT → STNOFF:SUMY:STAT

t ≈ 1 s SNL rf_states.st detects fault_stnoff != NO_ALARM:
→ fault_detected = 1
→ Transitions from s_on_cw → s_go_off state

t ≈ 1.0–1.1 s s_go_off Step 1: TUNESUB()
→ fba = 1 (beam abort forced from the software side)
→ runmode = TUNE

t ≈ 1.1–6 s s_go_off Steps 2–4: HVPS voltage setpoint → 0
→ rf_hvps_loop.st ramps HVPS voltage down to zero
→ ~5 s wait for capacitor bank to discharge

t ≈ 6 s s_go_off Steps 5–7:
→ hvpstrig = OFF (SCR supervisory enable removed via Remote I/O)
→ rfswitch = OFF (RFP drive enable removed)
→ intcomp = OFF

t ≈ 6–10 s ss rf_statesFF: Fault file capture
→ 11 signal channels × /dat/FAULT*_N (N = 1–15 circular)
→ Max 3.6 s for capture to complete

t > 10 s State machine in s_off; operator notified via EPICS alarm

9.2 Example: RF MPS PLC Trip — Cooling Water Overtemperature

This example illustrates a **slow equipment-protection fault** picked up by the RF MPS PLC (PLC-5/1771). No arc occurs — the PLC is the first actor. The cooling summary is not an input of the RF Local Display Panel logic [R27], so high voltage is removed through the RF cut and the collector-power cascade.

t = 0 s Cooling water return temperature sensor reads at scan interval:
→ Exceeds maximum setpoint continuously for N scan cycles
(debounce / confirmation prevents spurious trips)

t ≈ 10–30 ms RF MPS PLC scan cycle completes:
→ PLC program sets cooling overtemperature fault flag
→ AB summary fault set (Path A):
24 V line → RF Local Display Panel → fiber → Fast IC → AIM

t ≈ 10–30 ms AIM responds to the AB summary fault:
→ Asserts RF_FAULT on VXI P2 backplane (open-collector)
→ RFP module (Slot 4) monitors RF_FAULT
RF drive DAC output zeroed in hardware
RF power into klystron collapses
VXI Slot 5 MPS Shutoff module responds simultaneously (Path C):
→ Also asserts RF_FAULT on VXI P2 backplane
(redundant with AIM assertion above)

t ≈ 20–50 ms Collector-power cascade (§5):
→ PLC drops the collector summary
→ RF Local Display Panel logic removes SCR ENABLE
(fiber → HVPS controller, B118); HV output begins to collapse

t ≈ 5–50 μs AIM ISR fires:
→ Reads FISTAT latch register (Sass 2004 fix)
→ scanOnce({STN}:STN:AIM:MODU) scheduled

t ≈ 10–30 ms PLC writes permit bit OFF to Remote I/O output data table (Path B):
→ AB-6008 scanner reads at ~1 Hz
→ {STN}:STN:MPS:LTCH → MAJOR alarm set

t ≈ 50–500 ms EPICS propagates alarm:
→ {STN}:STNMPS:SUMY:LTCH.INPI = MAJOR
→ {STN}:STNOFF:SUMY:STAT.SEVR = MAJOR

t ≈ 1 s SNL rf_states.st detects fault_stnoff != NO_ALARM:
→ Transitions s_on_cw → s_go_off

t ≈ 1.0–6 s s_go_off orderly shutdown:
→ Step 1: fba = 1 (beam abort forced from the software side)
→ Steps 2–4: HVPS voltage setpoint → 0; ~5 s ramp
(HVPS already off through the cascade)
→ Step 5: hvpstrig = OFF (SCR supervisory enable removed)
→ Step 6: rfswitch = OFF (RF drive enable removed)

t ≈ 6–10 s ss rf_statesFF: Fault file capture
→ 11 signal channels → /dat/FAULT*_N files

t > 10 s State machine in s_off
→ EPICS operator display shows MPS trip: STN:MPS:LTCH = FAULT
→ No forced_fault latch set (cooling fault, not arc)
→ Auto-reset allowed once fault clears (cooling temp returns to normal)
→ Operator may allow auto-recovery or command manual restart

Key features of this type of trip: (1) The PLC is the first actor (~10–30 ms). (2) RF is cut through the AB summary and the AIM (with Slot 5 as a redundant RF cut); the HVPS follows through the collector-power cascade into the RF Local Display Panel logic, because the cooling summary is not an input of that logic. (3) No `forced_fault` latch is set, so SNL auto-reset may proceed after cooling fault clears. (4) The fault is identifiable by `{STN}:STN:MPS:LTCH = FAULT` with a normal (empty) `ARCLTDSTT` register.

9.3 Example: HVPS PLC Trip — High-Voltage Transformer Arc

This example illustrates a **HVPS internal fault** monitored by the SLC-500 HVPS PLC. The fault occurs inside the high-voltage transformer/oil tank at B514 — no RF waveguide arc, no MPS PLC involvement.

t = 0 ms Arc discharge inside the HV transformer tank (B514):
→ Arc detector sensor inside HV tank registers arc current surge
→ If arc is energetic enough: crowbar may self-fire via
analog crowbar trigger circuit (independent of HVPS PLC)

t ≈ 10–20 ms SLC-500 HVPS PLC scan cycle (B118):
→ Reads arc detector digital input:
Slot 6 IB16 input: C14.10 = HVPSXFORM:ARC:LTCH = FAULT
→ PLC ladder logic sets HVPS fault state
→ Slot 5 OX8 output 0 (24 V fault/enable) dropped
(supervisory thyristor-trigger enable removed on the COMMANDS bus)
→ Remote I/O output data table updated with fault status

t ≈ 10–20 ms Thyristor triggers inhibited by the SLC-500 supervisory path:
→ HV output begins to collapse (no gate drive on SCR thyristors)
→ HVPS STATUS fiber goes dark (it follows the 24 V fault/enable)
→ RF Local Display Panel receives it as the HVPS beam abort:
its logic latches the beam abort; its effect on SCR ENABLE
and the crowbar is set by the undocumented equations (O4)
→ The STATUS fiber does not reach the Fast IC or the AIM

t ≈ 100 ms–1 s EPICS alarm propagation:
→ AB-6008 scanner reads Remote I/O at ~1 Hz
→ {STN}:HVPSXFORM:ARC:LTCH = MAJOR
→ {STN}:HVPSSTN:SUMY:LTCH.INPG = MAJOR (transformer arc links in)
→ {STN}:HVPSOFF:SUMY:STAT = MAJOR
→ {STN}:STNOFF:SUMY:STAT.SEVR = MAJOR (via INPB)

t ≈ 1 s SNL rf_states.st detects fault_stnoff != NO_ALARM:
→ Transitions s_on_cw → s_go_off

t ≈ 1.0–1.1 s s_go_off Step 1: TUNESUB()
→ fba = 1 (beam abort forced from the software side)
→ runmode = TUNE

t ≈ 1.1–6 s s_go_off Steps 2–4: HVPS voltage setpoint → 0
→ rf_hvps_loop.st attempts HVPS voltage ramp down
(HVPS is already de-energized via the SLC-500 output;
ramp command is a no-op but the sequence must complete)
→ ~5 s wait

t ≈ 6 s s_go_off Steps 5–7:
→ hvpstrig = OFF (SCR supervisory enable again confirmed OFF)
→ rfswitch = OFF (RF drive enable removed)
→ intcomp = OFF

t ≈ 6–10 s ss rf_statesFF: Fault file capture
→ Waveform data captured to /dat/FAULT*_N files
NOTE: Fault occurred in HVPS tank at B514; signal content
in fault files shows the RF state at the time of HVPS collapse

t > 10 s State machine in s_off
→ Operator alarms: HVPSXFORM:ARC:LTCH and HVPSSTN:SUMY:LTCH
→ HVPS must be inspected before restart
→ No arc in RF waveguide: ARCLTDSTT = 0, FSTFLT = 0
→ No forced_fault latch (this is HVPS path, not Fast IC/AIM arc)
→ Recovery requires: clearing HVPS arc latch, verifying HVPS
oil/insulation condition, then normal restart sequence

Key features of this type of trip: (1) The SLC-500 HVPS PLC is the first actor (~10–20 ms); the RF MPS PLC and Fast IC are not involved. (2) HVPS shutdown is via the **SLC-500 supervisory output** (§7.6 path) — this is distinct from the SCR ENABLE fiber of the RF Local Display Panel. (3) The HVPS STATUS fiber goes dark with the 24 V fault/enable; the RF Local Display Panel receives it as the HVPS beam abort and drives its beam-abort output, but the fiber does not reach the Fast IC or the AIM. (4) **RF drive shutdown signal chain**: There is no documented hardware path from the SLC-500 HVPS PLC trip to the RFP module RF drive. The RF drive remains asserted at the software level until `s_go_off` Step 6 writes `rfswitch = OFF` at approximately t ≈ 6 s after the initial trip. During the intervening period the klystron cathode HV has already collapsed — the klystron produces no RF output even with drive signal still applied, because klystron amplification requires cathode high voltage. This is safe but the indirection must be understood. See §10.6 for the complete RF shutdown signal trace for this scenario. (5) Fault files capture the RF/IQA signal waveform at HVPS collapse time. (6) The `HVPSXFORM:ARC:LTCH` bit identifies the source; `ARCLTDSTT = 0` and `FSTFLT = 0` confirm no waveguide arc was involved.

9.4 Recovery Sequence

```
Operator observes fault via EPICS display
    → Checks fault files (/dat/FAULT*_N for latest fault slot)
    → Identifies fault channel (ARCLTDSTT bit, FSTFLT word, etc.)

Physical action:
    → Presses FAULT RESET on Fast Interlock Chassis front panel
      → Clears hardware arc latches in AIM module
      → Clears forced_fault latch
      → ARCLTDSTT bits cleared
      → STN:STN:FORCED:LTCH → OK → STNMPS:SUMY:LTCH clears

EPICS action:
    → fault_stnoff returns to NO_ALARM when all faults clear
    → Operator commands station to ON via EPICS display

SNL response:
    → s_off → s_go_park → s_go_tune → s_go_on_cw
    → HVPSONSUB():
        rfswitch = ON
        hvpstrig = ON  → SLC-500 (RIO adapter 1) output 0 on → thyristor triggers enabled
        aimon = ON     → AIM "HVPS_On" asserted → HVPS on request to the panel
    → RESET_BMABTSUB():
        fba = 0 → rba → rstf  (beam abort reset, arc hardware latches cleared)
    → Station climbs to ON_CW
```

9.5 Example: SPEAR MPS Permit Withdrawal — Cascade to Collector Overpower

This example traces the complete fault event when SPEAR MPS withdraws its beam permit (e.g., due to a beam loss or machine protection interlock condition). It illustrates the Slot 5 primary RF cut followed by the collector overpower cascade described in §5.

System state before fault: RF station ON_CW, klystron at rated output power, HVPS fully energized (~75 kV), normal beam circulating in SPEAR3.

`t = 0 ms` SPEAR MPS withdraws beam permit:
→ Slot 5 MPS Shutoff Module back-connector input drops

→ Slot 5 asserts RF_FAULT on VXI P2 backplane (hardware, < 1 μs)
→ RFP module (Slot 4) RF drive DAC → 0 (hardware, < 1 μs)
→ Klystron RF output: begins dropping to ~0 (~1–5 μs settling)

t ≈ 10 μs AIM Module (Slot 12) ISR fires:
→ STN:VXI:LTCH = MAJOR (EPICS Channel Access write)
→ STNPARK:SUMY:STAT alarm begins propagating through EPICS tree
At this instant
HVPS still fully energized at ~75 kV, ~10 A cathode current
RF output ≈ 0; cavity begins ring-down (~1 ms time constant)
ALL cathode beam power now depositing in klystron collector
P_collector ≈ P_cathode (= V_HV × I_cathode, typ. 500–750 kW)

t ≈ 10–20 ms RF MPS PLC (next scan cycle completes):
→ Reads KLYSCOLLPLC:POWER (RIO T4[41]): elevated above normal
(collector power ≈ full cathode input; well above operating value)
→ Internal comparator: measured power > KLYSCOLL:POWER:ULIM
→ PLC drops its collector summary → Path A activated:

t ≈ 15–25 ms Path A — RF Local Display Panel logic responds (μs scale):
→ SCR ENABLE fiber → HVPS controller (B118) goes dark
→ thyristor triggers inhibited
→ HVPS HV output begins to collapse
→ KLYSTRON CROWBAR fiber dark if the logic so specifies (O4)
→ crowbar discharges the residual capacitor energy
→ AB summary fiber → AIM → RF_FAULT (redundant RF cut confirmation)
Remote I/O latch: KLYSCOLL:POWER:LTCH (WL=16/WF=31/B=2) = MAJOR latched
(hardware latch confirming the PLC trip; propagates to EPICS)

t ≈ 30–50 ms HVPS STATUS fiber goes dark → RF Local Display Panel (HVPS beam abort)
[the panel's beam-abort output is already set — see §3.8]

t ≈ 100–300 ms EPICS Remote I/O data arrives at IOC:
→ {STN}:KLYSCOLL:POWER:LTCH propagates → KLYS:SUMY:LTCH = MAJOR
→ {STN}:STN:MPS:LTCH = MAJOR (Path B permit bit from PLC)
→ STN:ABSUMY:LTCH → STNOFF:SUMY:STAT = MAJOR
→ fault_stnoff ≠ NO_ALARM → SNL detects fault condition

t ≈ 1 s SNL s_go_off sequence begins:
→ Step 1: fba = 1 (beam abort forced from the software side)
→ Steps 2–4: HVPS ramp setpoint → 0 and hvpstrig=OFF
[no-op: HVPS already hardware-killed at ~20–25 ms]
→ Step 5: rfswitch = OFF (software confirms RF drive off;
was already 0 from Slot 5 at t = 0)
→ Step 6: Fault files captured (11 RF/IQA channels)
Station is in s_fault/s_off state.

EPICS fault signature:

PV	Value	Interpretation
`{STN}:STN:VXI:LTCH`	MAJOR	Slot 5 RF_FAULT seen by AIM ISR (root cause — timestamps ~10 μs after t=0)
`{STN}:STN:MPS:LTCH`	MAJOR	SPEAR MPS permit removed (root cause visible via Remote I/O Path B)

PV	Value	Interpretation
{STN}:KLYSCOLL:POWER:LTCH	MAJOR	Collector overpower trip in the PLC (cascade precipitating HVPS kill — timestamps ~15–25 ms after t=0)
{STN}:STN:AIM:ARCLTDSTT	0x000	No waveguide arc detected (all zeros — confirms not a cavity arc event)
{STN}:STN:ABSUMY:LTCH	MAJOR	AB summary fault from the RF MPS PLC (collector overpower)
{STN}:HVPS:VOLT	collapsing to 0 after ~25 ms	Thyristor triggers inhibited by the panel's SCR ENABLE

Key feature: The KLYSCOLL:POWER:LTCH flag appears as the hardware event that killed the HVPS — even though the root cause was SPEAR MPS. An operator must compare EPICS archiver timestamps: STN:MPS:LTCH timestamp precedes KLYSCOLL:POWER:LTCH by ~15–20 ms, confirming the cascade direction. With default 1 Hz archiver resolution, both may appear at the same timestamp; event-driven archiving is required for precise sequencing.

Recovery: Standard recovery sequence (§9.4). Both STN:MPS:LTCH and KLYSCOLL:POWER:LTCH must clear (SPEAR MPS must restore its permit; operator presses FAULT RESET) before the station can return to ON_CW.

---

10 Key Cross-Layer Interactions

10.1 AIM aimon ↔ SCR ENABLE Hardware Gate

The aimon write ({STN}:STN:AIM:MODU.HVPS = 1) sets the AIM "HVPS_On" output, which reaches the RF Local Display Panel as the HVPS on request. This is a **hardware interlock gate** — the panel's logic forms SCR ENABLE from it, so the HVPS controller in B118 cannot fire its thyristors unless the AIM has asserted it [R26], [R27]. This cross-link means:

- The SNL state machine must explicitly authorize HVPS energization (via HVPSONSUB)
- If the software is not in the ON transition sequence (e.g., the IOC is dead), AIM holds this output LOW and HVPS cannot be energized

In the fault path, aimon is NOT explicitly cleared in s_go_off — the hardware trip (SCR ENABLE removed by the panel logic + SLC-500 hvpstrig=OFF) has already shut down the HVPS. The explicit aimon=ON is only needed at startup.

10.2 forced_fault Anti-Auto-Reset

Without {STN}:STN:FORCED:LTCH, the auto-reset state (s_go_stn_reset) would retry reset_count times automatically. With forced_fault = MAJOR, the auto-reset loop exits immediately to s_off, requiring operator intervention. This prevents the HVPS from automatically re-energizing after a hard arc event — critical for SPEAR3 where arc repetition can damage cavity surfaces.

10.3 VXI Interlock Latch Register Fix (2004)

Before January 2004, VXI interlock bits were read from the AIM interrupt status register (ISR). EPICS processed the AIM:MODU record continuously, and each read cleared the ISR — a latched fault could be

invisible to the alarm system because it self-cleared between EPICS scans. Sass changed `rf_interlock_vxi.db` (rev 1.2, Jan 2004) to read from the AIM **fast interlock status/latch register**, which holds bits until the hardware FAULT RESET button is pressed. **This is a critical reliability fix** — without it, the software alarm chain would miss fast faults.

10.4 Dual SCR Enable Paths — Supervisory vs. Fast

The HVPS SCR bank has two independent enable paths that operate in different domains:

Path	Owner	Speed	Set by	Cleared by
RF Local Display Panel → SCR ENABLE fiber → trigger boards (B118)	Hardware interlock	μs	HVPS on request with no fault	Panel logic (O4)
SLC-500 output 0 → COMMANDS bus (B118)	Supervisory enable	~20 ms	SNL `hvpstrig=ON` in `HVPSONSUB`	SNL `hvpstrig=OFF` in `s_go_off`

The SLC-500 output is the **supervisory gate** — it deliberately enables the HVPS at the beginning of the startup sequence and is the controlled OFF path during orderly shutdown. The panel's SCR ENABLE fiber is the **fast-trip gate** — held lit during normal operation, dark on fault. Both must be satisfied simultaneously for the HVPS to produce HV output.

10.5 RF MPS PLC Three-Path Architecture — Defense in Depth

The RF MPS PLC is unique among the five actors in that a single fault produces consequences on three independent signal paths simultaneously (see §6.5 for full description). This is a cross-layer interaction worth noting explicitly:

Path	Layer	Immediate HVPS Effect	Delayed HVPS Effect (SNL)	RF Effect	Speed	Notes
A — MPS summaries → panel logic and AIM	Hardware interlock	**SCR ENABLE removed by the panel logic** for collector and waveguide pressure faults (fiber → HVPS controller, B118); crowbar per O4	Orderly `hvpstrig=OFF` (~1 s, HVPS already dead)	RF_FAULT via AIM	< 15 ms total	Primary HVPS shutdown path for the faults wired into the panel logic
B — Remote I/O permit bit	Software / EPICS	None	SNL orderly `s_go_off`: HVPS ramp + `hvpstrig=OFF`	SNL orderly shutdown	~1 s	Fault record, fault file, state machine bookkeeping; hardware already acted
C — MPS permit → VXI Slot 5	Hardware backplane	**None** — Slot 5 has no connection to the SCR ENABLE or crowbar fibers	Orderly HVPS shutdown via SNL `s_go_off` (~6 s, via `VXI:LTCH` alarm chain)	RF_FAULT via Slot 5	< 15 ms (RF cut)	Redundant RF cut; HVPS shutdown via SNL only

For SPEAR MPS permit and orbit interlock (Slot 5 direct inputs):

Input	Immediate HVPS Effect	Delayed HVPS Effect (SNL)	RF Effect	Speed
SPEAR MPS beam permit removal	**None** from Slot 5 — no panel path; **but collector overpower cascade fires Path A within ~20–30 ms (see §5)**	Primary: orderly SNL `s_go_off` (~6 s); **in practice: cascade hardware kill ~20–30 ms**	RF_FAULT via Slot 5	~5–15 ms (RF cut)
Orbit interlock trip	**None** from Slot 5 — no panel path; **but collector overpower cascade fires Path A within ~20–30 ms (see §5)**	Primary: orderly SNL `s_go_off` (~6 s); **in practice: cascade hardware kill ~20–30 ms**	RF_FAULT via Slot 5	~5–15 ms (RF cut)

The design consequence: for an RF MPS PLC trip on collector power or waveguide pressure, the immediate HVPS hardware shutdown is **owned by Path A** through the RF Local Display Panel logic; other MPS faults reach the HVPS through the collector-power cascade or the SNL. Paths B and C cannot produce immediate HVPS shutdown. If Path A were to fail (e.g., a panel fault), Path C would still cut RF drive, and the SNL would perform orderly HVPS shutdown ~6 s later, but there would be no immediate HVPS hardware kill.

For SPEAR MPS and orbit interlock signals, Slot 5 has **no connection** to the SCR ENABLE or crowbar fibers. There is no *primary* hardware HVPS kill from these signals. However, the RF drive cutoff invariably creates a cascade secondary trip: with drive = 0 and HVPS on, the klystron collector absorbs full cathode power, the RF MPS PLC drops its collector summary (Path A) within ~10–20 ms, and the panel logic removes SCR ENABLE within ~20–30 ms of the initial trip. In practice, HVPS hardware shutdown occurs within ~20–30 ms, not ~6 s. See §5 for the complete physics and signal chain, and §9.5 for an example fault timeline.

This asymmetry should be considered during upgrade architecture design: the new Interface Chassis interlock design must maintain an equivalent hardware-speed path from MPS signals to the HVPS SCR/crowbar circuit. Although the cascade (§5, §10.7) provides a de facto hardware HVPS kill within ~20–30 ms, it relies on the klystron physics and the RF MPS PLC — a **direct** wired path from SPEAR MPS permit removal to the HVPS SCR kill (bypassing the cascade dependency) is currently absent and should be evaluated for the upgrade.

10.6 RF Shutdown Path Following HVPS PLC Trip

Open question resolved. When the SLC-500 HVPS PLC trips and shuts off the HVPS, the RF drive is **not immediately cut** by any documented hardware signal from the HVPS PLC. The SLC-500 has no wired path to the Fast Interlock Chassis or the AIM; its only link to the RF side is the STATUS fiber, which ends on the RF Local Display Panel.

The complete RF shutdown signal trace for a HVPS PLC-only trip is:

`t = 0 ms` SLC-500 HVPS PLC detects internal fault (e.g., transformer arc):
→ Slot-5 output 0 (24 V fault/enable) drops
→ Thyristor triggers inhibited on the COMMANDS bus (B118)
→ HVPS HV output begins to collapse (SCR gates disabled)

`t ≈ 0 ms` HVPS STATUS fiber goes dark (it follows the 24 V fault/enable):
→ RF Local Display Panel receives the HVPS beam abort
→ The panel's beam-abort output is set; no documented path leads from it to the AIM or the RFP (O4)

t ≈ 0 ms KLYSTRON CATHODE VOLTAGE COLLAPSES:
→ Klystron cathode at approximately 0 V (or HVPS residual)
→ RF drive signal still present from RFP module (software unaware)
→ Klystron cannot amplify without cathode HV: RF output = ~0
→ Cavity fields decay on cavity time constant (~1 ms for SPEAR3)

t = 0 ms RF DRIVE SIGNAL PATH (VXI → klystron):
→ RFP module (VXI Slot 4) continues outputting RF drive DAC values
→ RF_FAULT NOT asserted (no AIM arc latch, no AB summary fault)
→ rfswitch PV still = ON

t ≈ 100 ms–1 s EPICS alarm propagation (Remote I/O → AB-6008 scanner → EPICS):
→ {STN}:HVPSXFORM:ARC:LTCH (or other HVPS fault latch) = MAJOR
→ {STN}:HVPSSTN:SUMY:LTCH propagates to HVPSOFF:SUMY:STAT
→ {STN}:STNOFF:SUMY:STAT.SEVR = MAJOR
→ fault_stnoff != NO_ALARM in rf_states.st

t ≈ 1 s SNL rf_states.st detects fault → enters s_go_off:
→ Step 1: fba = 1 (beam abort forced from the software side)
→ Steps 2–4: HVPS voltage setpoint → 0 (no-op; already collapsed)
→ Step 5: hvpstrig = OFF (SCR supervisory enable confirmed OFF)
→ Step 6: rfswitch = OFF
{STN}:STN:RFP:RFENABLE = 0
→ RFP module RF drive DAC output disabled
← RF DRIVE IS CUT HERE (t ≈ 6 s after initial trip)

Why the ~6 s latency is safe: A klystron is a velocity-modulation amplifier that requires cathode high voltage to produce RF output. Without cathode HV, even a full-amplitude drive signal produces no forward power. The cavity fields have completely decayed within ~1 ms of HV collapse. The ~6 s period between HVPS trip and `rfswitch = OFF` therefore does not produce any RF output and poses no equipment hazard — the klystron is a passive, inert component during this window.

No direct hardware RF kill path from HVPS PLC: Unlike an arc at the Fast IC (where the AIM asserts RF_FAULT to cut drive), the SLC-500 HVPS PLC trip path has no documented direct RF kill. Its STATUS fiber ends on the RF Local Display Panel, whose beam-abort output's endpoint is not documented (O4).

Possible indirect secondary path (not guaranteed): If the sudden HVPS voltage collapse causes reflected power to rise at the cavity junctions (forward power drops → VSWR increases), the RF MPS PLC may detect a reflected power trip condition and set its AB summary fault (Path A: fiber → AIM → RF_FAULT). This would cut RF drive in ~10–15 ms via AIM → VXI backplane. However, this secondary path is **not guaranteed**: if drive power was low before the trip, reflected power may stay below the MPS threshold.

RF shutdown timeline summary for HVPS PLC trip:

Time	Event	RF Drive State
t = 0	HVPS PLC trips; SCR ENABLE removed	Drive ON; klystron de-powered
t ≈ 0	HVPS HV collapses; klystron cathode at ~0 V	Drive ON (software unaware); RF output = 0
t ≈ 100 ms–1 s	EPICS alarms propagate via Remote I/O	Drive ON (SNL not yet triggered)
t ≈ 1 s	SNL s_go_off sequence begins	Drive ON (Steps 1–5 in progress)
t ≈ 6 s	**s_go_off Step 6: rfswitch = OFF**	**Drive OFF — RF drive cut**

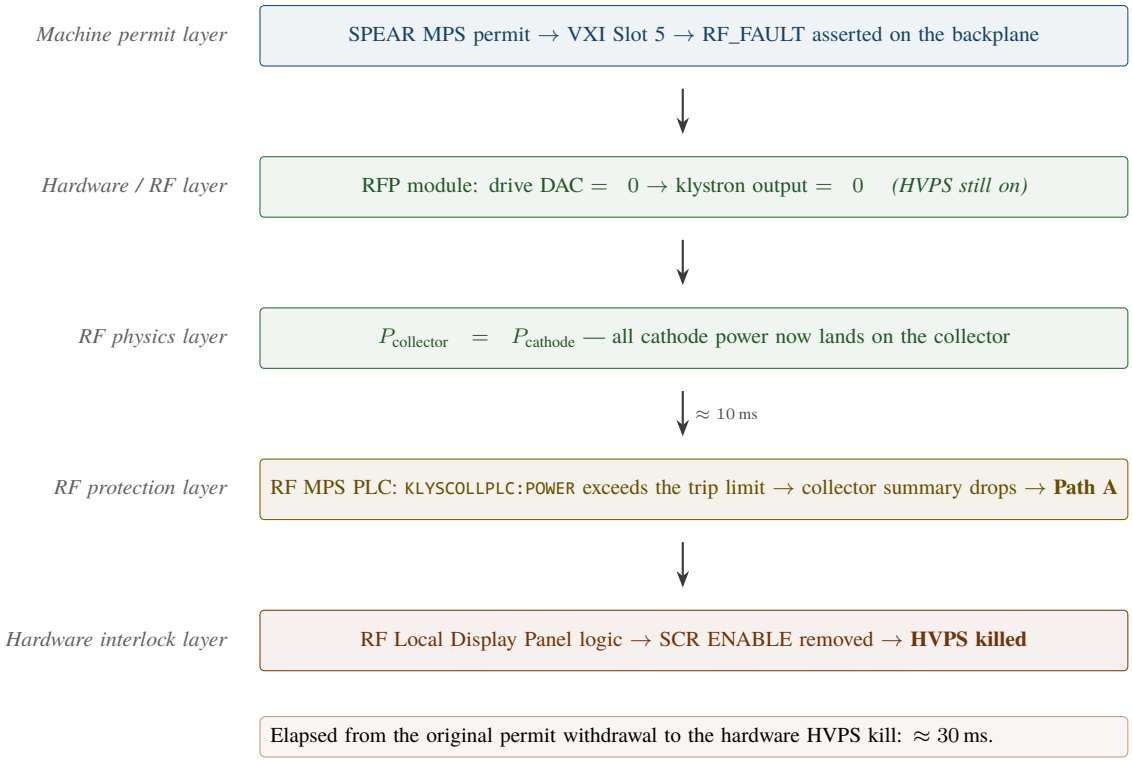


Figure 8: Permit layers. The machine permit and personnel permit chains are separate all the way down.

Design implication for the upgrade: The upgrade Interface Chassis interlock design should include a direct hardware path from HVPS enable removal (or HVPS PLC fault output) to the LLRF9 RF enable input. This would eliminate the ~6 s software-dependent window and make the RF shutdown timing visible and auditable in hardware — consistent with the defense-in-depth philosophy of the rest of the interlock architecture.

10.7 Cascade Cross-Layer Interaction: RF Drive Cutoff → Secondary Trip Chain

The collector overpower cascade (§5) is a cross-layer interaction in which an event that originates in the machine-protection / permit layer (SPEAR MPS permit withdrawal → Slot 5 RF cut) propagates back into the hardware protection layer (RF MPS PLC collector summary → RF Local Display Panel logic), producing an immediate hardware-speed HVPS kill.

This behavior is consistent with the defense-in-depth design principle: the RF MPS PLC independently monitors a physical parameter (collector power) that is deterministically correlated with the RF drive state. It does not need to ”know” that a SPEAR MPS event occurred — it simply responds to an out-of-bounds physical condition.

Cross-layer cascade path (Figure 8):

Three key design observations:

1. **The cascade is deterministic** for normally-operating klystrons. RF drive cutoff with HVPS energized will always drive the collector to absorb full cathode power. As long as the trip limit (`KLYSCOLL:POWER:ULIM`) is set below the full-cathode-power value (which it must be to protect the klystron), the PLC will drop its collector summary within one scan cycle.

2. **Path A is the hardware HVPS kill for collector overpower** — whether triggered by a primary RF MPS PLC protection event or by this cascade, the collector summary into the RF Local Display Panel logic is the mechanism. Any detected collector overpower therefore removes SCR ENABLE in hardware, regardless of how the overpower arose.
3. **Fault diagnosis transparency** — the cascade produces a fault signature dominated by `KLYSCOLL:POWER:LTCH` (collector overpower), while the actual root cause is visible in `STN:MPS:LTCH` and/or `STN:VXI:LTCH`. The fault analysis procedure (§11.6) should explicitly check for this cascade pattern when a collector overpower latch coincides with a machine permit withdrawal:
   - `STN:VXI:LTCH` timestamps $\ll$ `KLYSCOLL:POWER:LTCH` timestamps $\rightarrow$ cascade (root cause is the permit withdrawal)
   - `KLYSCOLL:POWER:LTCH` timestamps with no preceding permit change $\rightarrow$ primary collector overpower event (possible klystron efficiency issue)

Upgrade architecture note: The cascade provides a *de facto* hardware HVPS kill for SPEAR MPS and orbit interlock trips via an indirect physics path. The upgrade Interface Chassis design should consider whether to implement a *direct* hardware path (permit removal $\rightarrow$ HVPS SCR kill, without depending on the klystron physics cascade), which would be faster (~1 ms vs. ~20–30 ms), more transparent in fault records, and functional even at low RF power levels where the cascade may not fire.

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11 Fault Data Availability and Analysis

When a fault event occurs, multiple data sources capture pre- and post-event waveforms at different speeds and with different signal content. This section describes where each data source is stored, how to access it, and provides a step-by-step fault analysis procedure.

11.1 Overview of Fault Data Sources

Source	What It Captures	Time Resolution	Latency to Availability	Storage Location
AIM Hardware History Buffer	12 arc channel voltages + HVPS voltage (continuous ADC ring buffer)	~μs per sample	Immediately (freezes on fault)	VXI AIM module on-board memory; exported to IOC `/dat/aimHist.dat`
SNL Fault Files (`/dat/FAULT*_N`)	11 RF/IQA signal channels (I/Q waveforms, amplitude waveforms)	~1 ms per sample	~6–10 s post-fault	VxWorks virtual disk `/dat/` on VXI IOC
B118 Oscilloscope (4-channel)	HVPS DC voltage, HVPS DC current, inductor T2 sawtooth voltage, transformer T1 AC current	Oscilloscope sweep rate (operator-set; typical 10–100 ms/div)	Continuous display; capture on trigger	Local oscilloscope (memory, USB, or printout)
EPICS Channel Archiver	All EPICS PVs — alarm states, analog readbacks, timestamps, state machine transitions	1 Hz nominal; event-driven on PV change	Always available; rolling multi-year history	EPICS archiver server (see §11.5)

11.2 AIM Hardware History Buffer

What it contains: The Fast Interlock Chassis (340-308) onboard fast ADC continuously digitizes all 12 arc channel voltages and the HVPS DC voltage into the AIM module's 512 KB ring buffer (`HISBUF`, A24 address space, offset `0x0038`). On fault, the `ADCCTL` bit freezes the buffer, preserving the pre-fault arc waveforms.

Where stored: VXI AIM module on-board memory. The device driver (`devP2RfAim.c`) reads arc voltage peaks via the `ARCVOL` register and crate voltage via `CRTVOL`. The history data is exported to `/dat/aimHist.dat` on the IOC. Note: buffer readout is **gated** — if `FISTAT.HVPSON = 0` (HVPS off at fault time), the driver skips arc voltage readback.

EPICS readback PVs (immediately available post-fault):

PV	Content
`{STN}:STN:AIM:ARCCURSTT`	12-bit real-time arc channel status (clears when arc ends)
`{STN}:STN:AIM:ARCLTDSTT`	12-bit latched arc channel status (holds until FAULT RESET button pressed)
`{STN}:STN:AIM:FSTFLT`	Fast IC aggregated fast fault status word
`{STN}:STN:AIM:INTSTATE`	AIM overall interlock state

Interpreting `ARCLTDSTT` bit pattern:

Bits	Sensor Location	Notes
0–3	Four cavity waveguide window arc sensors	Cavity 1–4 in numerical order
4	Klystron output window arc sensor	High-energy arc in klystron output waveguide
5	Main circulator arc sensor	Arc at circulator assembly
6–11	Spare channels (not connected in SPEAR3)	Should always be 0
All zeros	**No waveguide arc detected**	Fault originated from HVPS, MPS PLC, or other non-arc source

11.3 SNL Fault Files (`/dat/FAULT*_N`)

What they contain: 11 pre-fault RF and IQA signal channels captured from hardware signal memory buffers in the RFP and IQA VXI modules. These represent the RF state at the moment of fault.

Where stored: VxWorks virtual disk `/dat/` on the VXI IOC in B132. Files use the naming convention `FAULT<channel>_<slot>` where `<slot>` is 1–15 (circular buffer, `NUMFAULTS=15`). The current slot number is tracked as EPICS PV `{STN}:STN:FAULT:NUM`.

File channel list (in capture order for SPEAR3 `#else` branch of `#ifdef CF2`):

Slot Index	File Pattern	Signal
0	`FAULTRfpSI_N`	RFP Signal I (cavity forward, I component)
1	`FAULTRfpSQ_N`	RFP Signal Q (cavity forward, Q component)
2	`FAULTRfpCI_N`	RFP Cavity I (cavity voltage, I component)
3	`FAULTRfpCQ_N`	RFP Cavity Q (cavity voltage, Q component)
4	`FAULTIqa1Amp_N`	IQA1 amplitude waveform
5	`FAULTIqa2Amp_N`	IQA2 amplitude waveform
6	`FAULTGvf_N`	GVF signal (PEP-II heritage; inactive in SPEAR3)
7	`FAULTAim_N`	AIM arc/interlock status snapshot
8	`FAULTCmbI_N`	Combiner I (PEP-II; inactive in SPEAR3)
9	`FAULTIqa3Amp_N`	IQA3 amplitude waveform
10	`FAULTCmbQ_N`	Combiner Q (PEP-II; inactive in SPEAR3)

Access procedure:

1. Check `{STN}:STN:FAULT:NUM` EPICS PV to find the most recent fault slot number N
2. Log in to the VxWorks IOC shell (telnet to B132 VXI CPU) or access via NFS/FTP
3. List `/dat/FAULT*_N` (substitute the slot number)
4. Transfer files to a workstation for analysis (binary format; see §11.6 for reading)

11.4 B118 Oscilloscope — Four HVPS Monitor Channels

Location: Building B118, HVPS Hoffman Box equipment area. An oscilloscope is permanently installed to monitor 4 analog signals from the HVPS system.

The four channels are physical analog signals routed from the HVPS cabinet to BNC inputs on the oscilloscope via shielded cables. The oscilloscope is a standalone instrument; it is **not** connected to EPICS. The signal identities are:

Channel	Signal	Source Hardware	Expected Waveform Pattern	Diagnostic Value
CH1	HVPS DC Output Voltage	Resistive voltage divider (≈10,000:1) from HVPS DC bus	Smooth DC at ≈ −72 to −75 kV typical; slight 12-pulse ripple (~6.9% P-P)	Crowbar: instant collapse to ~0 V; SCR disable: decay over ~100 ms; setpoint changes visible as slow ramp
CH2	HVPS DC Output Current	Danfysik DC-CT (Hall-effect current transducer)	DC at ~22 A nominal	Arc event: current spike; crowbar: current spike then zero; HVPS trip: current decay
CH3	Inductor 2 (T2) Sawtooth Voltage	SCR firing circuit monitor winding on T2	**Bipolar asymmetric sawtooth** (inverted direction vs. theoretical model — this is normal for real SPEAR3)	Waveform disappears when SCR firing stops; asymmetry changes indicate firing angle / regulation shifts; absence indicates HVPS off

Channel	Signal	Source Hardware	Expected Waveform Pattern	Diagnostic Value
CH4	Transformer 1 (T1) AC Phase Current	Current transducer on T1 primary winding	**Bipolar asymmetric square pulses** with commutation spikes (not sinusoidal — characteristic of 12-pulse discrete switching)	Dropout indicates HVPS de-energized; asymmetric pulses indicate phase balance issue; commutation spike amplitude indicates SCR health

Accessing data for fault analysis:

- **Field capture**: Use the oscilloscope's single-shot trigger or freeze function at the time of fault; export via USB/Ethernet or photograph the screen. The scope is a standalone instrument — no EPICS waveform interface exists for this unit.

11.5 EPICS Channel Archiver

What it contains: The SLAC EPICS Channel Archiver continuously records all EPICS PVs with timestamps. This provides the complete alarm and measurement history for all PVs in the RF station, typically with years of rolling history.

Access methods:

1. **EPICS Strip Chart / Archive Viewer GUI**: Open the Strip Chart tool from any EPICS operator display. Enter PV name(s) and select the time window around the fault event.
2. **Channel Archiver web interface**: Contact SLAC SSRL controls group for the archiver server hostname and web URL.
3. **Python (Archiver Appliance REST API)**:

```
import requests
pv = "SRF1:STN:AIM:ARCLTDSTT"  # substitute actual station prefix
t0 = "2026-04-10T02:00:00Z"      # start time (UTC ISO-8601)
t1 = "2026-04-10T03:00:00Z"      # end time
url = f"http://<archiver-host>/retrieval/data/getData.json?pv={pv}&from={t0}&to={t1}"
data = requests.get(url).json()
```

Key PVs to retrieve for fault post-mortem:

Category	PV	What to Look For
Arc fault	`{STN}:STN:AIM:ARCLTDSTT`	Which bit(s) set at fault time?
Fast IC fault	`{STN}:STN:AIM:FSTFLT`	Fast IC fault word at trip instant
MPS PLC	`{STN}:STN:MPS:LTCH`	Did MPS PLC trip independently of arc?
HVPS summary	`{STN}:HVPSSTN:SUMY:LTCH`	Which HVPS sub-fault latched?
Transformer arc	`{STN}:HVPSXFORM:ARC:LTCH`	HV transformer internal arc event?

Category	PV	What to Look For
Crowbar	`{STN}:HVPS:CROWBAR:LTCH`	Crowbar activated?
Overvoltage	`{STN}:HVPS:VOLT:LTCH`	DC output overvoltage?
HVPS voltage	`{STN}:HVPS:VOLT`	Pre-fault and collapse waveform (1 Hz; limited time resolution)
Station state	`{STN}:STN:STATE`	State machine transitions: ON_CW → s_go_off → OFF
Fault slot	`{STN}:STN:FAULT:NUM`	Fault file slot number — use to locate `/dat/FAULT*_N` files
Forced fault	`{STN}:STN:FORCED:LTCH`	Hard arc latch — requires manual FAULT RESET to clear

11.6 Step-by-Step Fault Analysis Procedure

11.6.1 Immediate Response (within minutes of fault)

1. **Record exact UTC timestamp** from the EPICS alarm notification or operator display
2. **Read the EPICS alarm display** on the EDM operator panel — identify which latches are set:
   - `ARCLTDSTT` $\neq$ `0` → Arc fault; note which bit(s)
   - `HVPSXFORM:ARC:LTCH` set → HV transformer internal arc
   - `HVPS:CROWBAR:LTCH` set → Crowbar fired
   - `STN:MPS:LTCH` set → MPS PLC trip (cooling, vacuum, or power)
   - `STN:FORCED:LTCH` set → Hard arc; manual FAULT RESET required before recovery
3. **Note fault slot number** from `{STN}:STN:FAULT:NUM` — needed to retrieve fault files
4. **Go to B118** and check the oscilloscope display for the fault waveforms:
   - Was CH1 a sudden drop (crowbar) or a slow decay (SCR disable)?
   - Did CH4 drop at the same instant as CH1, or before?
5. **Do NOT press FAULT RESET** on the Fast Interlock Chassis front panel until the cause is understood — pressing it clears the arc latch evidence

11.6.2 Detailed Analysis (after initial survey)

1. **Retrieve fault files** from IOC: `telnet <B132-IOC-hostname>`, then check `/dat/FAULT*_N` for the slot noted above
2. **Retrieve EPICS archiver history** for the key PVs listed in §11.5 in a $\pm$5 minute window around the fault timestamp
3. **Plot fault file data**. The files are binary with a short header followed by raw ADC samples as 16-bit integers or 32-bit floats (format is hardware-dependent; consult `devP2RfAim.c` for AIM fault file format or the RFP/IQA driver for RF signal file format). In Python:

```
import numpy as np
data = np.fromfile('/dat/FAULTRfpSI_1', dtype=np.int16)  # example
```

4. **Review the B118 oscilloscope capture** (see §11.4): note whether CH1 (DC voltage) shows an abrupt collapse (crowbar, < 1 μs) or a gradual decay (~100 ms for SCR disable), and whether CH4 (T1 AC current) dropped simultaneously with or prior to CH1

5. **Determine the initiating event** — use EPICS timestamp ordering (1 Hz resolution for Remote I/O PVs) to sequence events among HVPS faults; use the arc channel bit pattern for hardware-speed (< 1 μs) faults

11.6.3 Fault Categorization Guide

Observed Indicators	Most Likely Initiating Event	First Actor
`ARCLTDSTT` bits 0–3 set	Cavity waveguide window arc (cavity N = bit N)	Fast IC (< 1 μs)
`ARCLTDSTT` bit 4 set	Klystron output window arc	Fast IC (< 1 μs)
`ARCLTDSTT` bit 5 set	Main circulator arc	Fast IC (< 1 μs)
`ARCLTDSTT = 0`, `HVPSXFORM:ARC:LTCH` set	HV transformer internal arc	SLC-500 HVPS PLC (~10–20 ms)
`ARCLTDSTT = 0`, `HVPS:CROWBAR:LTCH` set, no arc latch	HVPS overvoltage → crowbar self-fired; OR analog crowbar threshold exceeded	Crowbar analog circuit (< 1 μs)
`ARCLTDSTT = 0`, `STN:MPS:LTCH` set	RF MPS PLC trip (cooling water over-temp, vacuum excursion, or overpower)	RF MPS PLC (~10 ms)
`ARCLTDSTT = 0`, `HVPSOIL:TEMP:LTCH` set	Oil overtemperature (slow thermal fault)	SLC-500 HVPS PLC (~10–20 ms)
`ARCLTDSTT = 0`, `HVPS:VOLT:LTCH` set	HVPS DC output overvoltage	SLC-500 HVPS PLC (~10–20 ms)
CH3 waveform entirely absent (oscilloscope)	SCR firing stopped — contactor open, or full SCR fault	Hardware, timing depends on cause
CH4 waveform asymmetric or one phase missing	Phase imbalance in primary; possible SCR failure or phase loss	SLC-500 or AC protection
CH1 voltage collapse: instant step to ~0 V	Crowbar fired	Crowbar circuit (trigger signal < 1 μs; the crowbar then **conducts** $\approx$ **10** μ**s** later, and the primary current is interrupted in **4–8 ms** — SLAC-PUB-7591)
CH1 voltage collapse: exponential decay over ~100 ms	SCR gate drive disabled (supervisory path)	SLC-500 relay (~10–20 ms)

Sources: §3.9 (AIM history buffer and SNL fault files); §7.5 (HVPS PV list); §10.6 (RF shutdown path); `rfApp/Db/rf_hvps.db` (HVPS scalar PV definitions); `llrf/llrf9/iGp/matlab/llrf/read_waveforms.m` (LabCA MATLAB access pattern); `iocBoot/b132-iocrf/Tables/device.tbl` (KSC2961 GPIB controller registration).

A EPICS PV Quick Reference

PV	Description	Normal State
{STN}:STNOFF:SUMY:STAT.SEVR	**Master trip wire** (SNL fault_stnoff)	NO_ALARM
{STN}:STN:AIM:MODU.HVPS	AIM HVPS permissive (aimon)	1 when ON
{STN}:STN:AIM:FRCBMABT	Force Beam Abort (fba)	0 when ON
{STN}:STN:AIM:MODU.RSTF	Reset AIM Faults (rstf)	— (write only)
{STN}:STN:AIM:MODU.RBA	Reset Beam Abort (rba)	— (write only)
{STN}:STN:AIM:ARCLTDSTT	Arc latched status (12-bit word)	0 = no arc
{STN}:STN:AIM:ARCCURSTT	Arc current status (12-bit word)	0 = no arc
{STN}:STN:AIM:FSTFLT	Fast fault status from Fast IC	0 = OK
{STN}:STN:VXI:LTCH	VXI interlock latch (from AIM latch register)	OK
{STN}:STN:FORCED:LTCH	Forced fault latch (blocks auto-reset)	OK
{STN}:STN:MPS:LTCH	RF MPS PLC permit status	OK
{STN}:HVPSSCR:ON:CTRL	HVPS SCR enable switch (hvpstrig)	1 when ON
{STN}:STN:RFP:RFENABLE	RF drive enable (rfswitch)	1 when ON
{STN}:HVPS:CROWBAR:LTCH	HVPS crowbar fired latch	OK
{STN}:HVPS:VOLT:LTCH	HVPS overvoltage latch	OK
{STN}:HVPSOIL:LEVEL:LTCH	HVPS oil level latch	OK
{STN}:HVPSOIL:TEMP:LTCH	HVPS oil temperature latch	OK
{STN}:HVPSXFORM:ARC:LTCH	Transformer arc latch	OK
{STN}:HVPS:PANIC:LTCH	Emergency Off latch	OK
{STN}:HVPSOFF:SUMY:STAT	HVPS OFF status summary	NO_ALARM
{STN}:HVPSSTN:SUMY:LTCH	HVPS station fault summary	NO_ALARM
{STN}:STNMPS:SUMY:LTCH	Station MPS summary (SPEAR3)	NO_ALARM

B Source Reference Index

All findings in this document were derived from direct analysis of the following source files. References are cited throughout as **[Rn]**.

B.1 Design Documents

Ref	File	Description
[R1]	Designs/tex/docL-legacy-architecture/L_legacy_system_architecture.pdf	SPEAR3 RF System Legacy Architecture (v2.5+) — primary system reference

B.2 Primary Source Code

Ref	File	Location	Lines	Role
[R2]	rf_states.st,v	rfApp/src/seq/	2,227	Master SNL state machine — all fault detection and response logic
[R3]	devP2RfAim.c,v	rfApp/src/db/	1,982	AIM module device support — arc channels, ISR, BATS, fault files
[R4]	devABSLCDCM.c,v	allen Bradley/allen Bradley App/SLCDCM Src/	563	SLC-500 Remote I/O device support — HVPS supervisory communication
[R5]	p2RfCf2Def.h,v	rfApp/src/db/	403	CF2 register definitions (PEP-II heritage, slot 5 reference)
[R6]	p2RfAimDef.h,v	rfApp/src/db/	—	AIM register definitions (TRIPLVL, HISBUF, ARCVOL, FISTAT, HVPSON)

B.3 EPICS Database Files

Ref	File	Location	Role
[R7]	aim.db,v	rfApp/Db/	AIM module PV definitions — arc status, interlock state, control outputs
[R8]	rf_digital_hvps.db,v	rfApp/Db/	HVPS digital I/O template — all HVPS status/latch bits from SLC-500
[R9]	rf_digital_All.substitutions,v	rfApp/Db/	Instantiates HVPS digital I/O with Remote I/O word/bit assignments
[R10]	rf_sumy_hvps.db,v	rfApp/Db/	HVPS alarm aggregation tree
[R11]	rf_sumy_stn.db,v	rfApp/Db/	Station-level alarm aggregation — STNOFF/STNON/STNMPS trees
[R12]	rf_sumy_stn_spr.db,v	rfApp/Db/	SPEAR3-specific additions to station summary (STNMPS, external MPS)
[R13]	rf_interlock.db,v	rfApp/Db/	DCM-based hardware interlock channels (MPS wiring)
[R14]	rf_interlock_vxi.db,v	rfApp/Db/	VXI-based interlock channels — AIM latch register bits
[R15]	rf_analog_All.substitutions,v	rfApp/Db/	Analog channel assignments (HVPS volts, current, temperature)
[R16]	rf_hvps.db,v	rfApp/Db/	HVPS supervisory records — voltage setpoints, SCR control, oil temp
[R17]	crat_vxi_13slot.template,v	rfApp/Db/	VXI crate slot labeling — confirms slot 5 = ”MPS Shutoff”
[R18]	rf_vxi_modules_All.substitutions,v	rfApp/Db/	VXI module DB loading — cf2.db assigned to slot 5 (PEP-II artifact)

B.4 Technical Notes (Code Analysis Series)

Ref	File	Location	Content
[R19]	`00-executive-summary.md`	`codeReviewTechnicalNotes/`	System overview, slot 5 clarification
[R20]	`01-file-inventory.md`	`codeReviewTechnicalNotes/`	File classification — CF2 marked PEP-II only
[R21]	`03-vxi-device-support.md`	`codeReviewTechnicalNotes/`	AIM: arc channels, fast interlock chain, BATS, fault files
[R22]	`05-snl-state-machines.md`	`codeReviewTechnicalNotes/`	Complete 23-state SNL architecture documentation
[R23]	`06-plc-stepper-motors.md`	`codeReviewTechnicalNotes/`	PLC topology: 3 Remote I/O adapters, SLC-500 HVPS functions
[R24]	Switchgear and contactor drawings	`hvps/documentation/switchgear/`	GP-439-704-02-C1 (schematic), ID-308-801-06-C1 (connection wiring), GP-308-500-01-R3 (original LBL design, relay legend), Ross 713203 E-1 (HQ3 contactor and HCA-1-A driver). All verified by direct reading, September 2026
[R25]	`HoffmanBoxPPSWiring.docx`	`pps/`	J. Sebek — original terminal-by-terminal trace of the B118 Hoffman Box PPS wiring, switchgear theory of operation, and the K4/RR label-swap correction
[R26]	`LLRFDocumentationNotesR2.docx`	`llrf/documentation/`	J. Sebek — Remote I/O topology; the fibers between the AIM, the Fast Interlock Chassis, the RF Local Display Panel and the HVPS controller; the panel's cross-connects
[R27]	`fiberOpticCableSignalControlRev3.docx`	`llrf/documentation/`	J. Sebek — the three fibers between the LLRF system and the HVPS controller, their light states and the inputs of the panel logic
[R28]	`controllerFiberOpticConnections.docx`	`hvps/architecture/designNotes/`	J. Sebek — HVPS controller side: slot-5 output 0 (24 V fault/enable) on the COMMANDS bus of the trigger interconnect boards
[R29]	RF Local Display Panel drawings	`llrf/documentation/localPanel/`	DL-340-310-00, front and rear panels SI-340-310-05 and -02, schematic SD-340-311-01 (GAL22V10 logic, fiber transceivers, beam-abort BNC J6)
[R30]	Wiring diagrams WD-340-330-29 and -02	`llrf/documentation/mpsWiringDiagrams/`	Control fiber-optic cables of the RF Local Display Panel; cross-connect X530 to its connectors J2 and J3
[R31]	`SP3_writeup_V1d2.pdf`	`llrf/documentation/legacyArchitecture/`	J. Dusatko, SPEAR3 LLRF system description, v1.2 — the AIM's six control signals, set by register bits and its on-board fault logic